How Much Targeted Investment Makes a Model First? Cost-to-First on Two Public Multi-Task Leaderboards under Different Aggregation Rules

Anonymous authors
Paper under double-blind review

Abstract

A multi-task leaderboard gives a developer a reason to train a chosen model on data related to some tasks. For two fixed public score tables we ask how much targeted investment makes a target model first, and how this depends on the aggregation rule. The target raises its task scores within a per-task cap while rivals stay fixed; we solve exactly for the fewest tasks touched (K), the fewest score points (L_1) and the smallest largest single-task rise (L_∞) under five score rules, plus Borda, plurality and a Condorcet requirement on HELM and a top-500 frame. The tables are HELM MMLU (79 models, 57 subjects) and the Open LLM Leaderboard (4,496 models, 6 tasks). We prove that no universal ordering of the mean, median and minimum exists. (i) With unrestricted headroom the mean needs the most raw score points but not always the most tasks, and not the most points in standard-deviation units. (ii) With at most $+10$ points per task, the mean can put 29 of 78 HELM targets first (37%, resampling range 26–49%; 14% at $+5$, 72% at $+20$) and 113 of 4,495 Open LLM Leaderboard targets (2.5%, 2.0–3.0%; 0.3% and 18%). Whether the minimum rule is more reachable than the mean at $+10$ differs between the tables, but not only because of the table: matching rivals or tasks keeps each sign in most random subsets, while a fixed mean-selected frame and a task-IQR cap reverse it. We name no single cause. (iii) Lowest-score-first greedy already attains the optimal number of tasks at the ceiling; random order needs 1.8, 1.1 and 4.2 times the optimum on HELM under the mean, median and minimum. All numbers describe fixed snapshots, not what training achieves.

1 Introduction

Multi-task benchmarks and leaderboards (MMLU, HELM, the Open LLM Leaderboard) rank models by aggregating scores on many sub-datasets, typically with a mean. A leaderboard position influences adoption, publication and funding, so developers have a reason to act strategically, for instance by putting data related to some sub-datasets into a specific model's training in order to raise its scores there. Prior work shows that training on benchmark-related data can inflate scores (Zhou et al., 2023; Yang et al., 2023), that leaderboard practices can distort rankings (Singh et al., 2025), and that incentives of this kind can be analysed with game-theoretic models (Chen et al., 2026). What is missing is a direct, computable answer to a concrete question: *on a given public leaderboard, how much targeted investment does it take for a chosen model to become first, and how does that depend on the aggregation rule?*

We answer it with an explicit model (Section 3): the target may raise its score on any task up to a per-task cap while all rivals stay fixed, and "first" means strictly above every rival by a small margin. We measure investment by the number of tasks touched K, the total points L_1 and the largest single-task rise L_∞, each minimized exactly, and treat the cap as a scenario, never as an estimate of achievable training gains. We compare five score rules (mean, median, minimum, geometric and harmonic means) and three ordinal rules (Borda, plurality, Condorcet winner) on every model of two public leaderboards (score rules on all models; ordinal rules on HELM and a fixed top-500 frame of the

Open LLM Leaderboard, with the Condorcet requirement an upper bound for Condorcet-consistent rules).

Contributions and scope of the claims.

1. *Theory (elementary, with proofs in Appendix C).* Closed forms for the mean, minimum and median (the cardinal analogues of margin-of-victory computations), a threshold equivalence between L_∞ and per-task gain caps (for any rule with a closed first-placing set), and constructed examples proving that **no universal ordering of the mean, median and minimum rules holds** by K or L_1 (Section 3). The empirical census below is the main contribution.
2. *Measurement.* Exact, solver-validated costs to be first, with ranges from resampling the targets, for all models of HELM MMLU (79 models, 57 subjects) and the Open LLM Leaderboard (4,496 models, 6 tasks), with infeasibility reported explicitly (Section 5). The answer is conditional on frame, cap and measure: raw points needed are largest under the mean; the number of tasks needed and, under a cap, whether first place is reachable at all, depend on the frame, the cap and the unit.
3. *Controls and sensitivity.* Targeting beats random task order and uniform improvement, but a lowest-score-first greedy order already attains the optimal task count at the ceiling cap, so the optimiser's value lies in points and in capped settings; matched rival and task counts, task resampling and score noise show which comparisons are stable (Section 5, Appendix).

All numbers are cost-to-first computations on fixed score snapshots; they describe those censuses and say nothing about what training on related data would actually deliver, nor about any developer's behaviour.

2 Related work

Multi-task benchmarks as voting. Zhang & Hardt (2024) and Lanctot et al. (2023) view tasks as voters and models as candidates; the former separates *cardinal* benchmarks (aggregating scores) from *ordinal* ones (aggregating per-task rankings), applies Arrow's theorem to ordinal ones and measures sensitivity to irrelevant changes; the latter proposes voting rules as evaluation frameworks. These works ask how a rule responds to changes in the set of models or tasks; we ask what it costs a *single* model to win. Dehghani et al. (2021) show that the choice of tasks can reorder methods.

Manipulation in social choice. Minimum-cost election manipulation is classical in computational social choice (bribery, shift bribery, margin of victory; Faliszewski et al., 2006; Brandt et al., 2016). That literature is ordinal and studies computational complexity. Ours is the cardinal counterpart: the target's task scores move continuously inside a box, rivals are fixed, and the rules are score aggregators (mean, median, minimum, generalized means) and, for comparison, ordinal rules (Borda, plurality, a Condorcet requirement). The closed forms for the mean, minimum and odd median are elementary (they are the cardinal analogue of a margin-of-victory computation), so what we add is the exact computation on real score tables for every model, the reversals of the rule ordering, and the dependence on cap and frame.

Strategic behaviour and contamination. Chen et al. (2026) model benchmarking as a designer–developer game: each developer has a latent capability, a scalar score and a scalar effort, and is paid by rank. Any pure equilibrium keeps the capability order, the only candidate is zero effort, and it exists iff the cost of the *just-overtake* effort (to pass the model directly above) is at least the adjacent reward gap (their Prop. 4.5, Thm. 4.6); under winner-take-all the binding pair is second against first. Their overtaking cost is the single-score counterpart of our cost-to-first, and tune-before-test raises it. Multiple tasks enter only through one fixed linear weight on the latent capability, with no per-task effort and no choice of aggregation rule, so their results order no rules; the rule, task-by-task and frame dependence measured here lie outside them. Their evidence is a case study of five Qwen2.5 sizes; ours is every model of two leaderboards under fixed rivals, with no equilibrium claim. Singh et al. (2025) document distortions of the Chatbot Arena leaderboard (private testing, sampling). Zhou et al. (2023) and Yang et al. (2023) show that training on benchmark-related data inflates scores and can evade decontamination. These motivate our investment model but concern

measured effects of leakage; we do not measure achievable gains and treat the per-task gain cap as a scenario.

Benchmarks used. We use per-subject MMLU (Hendrycks et al., 2020) accuracies from HELM (Liang et al., 2022) and the six tasks of the Open LLM Leaderboard (v2; IFEval, BBH Suzgun et al., 2022, MATH Lvl 5, GPQA, MuSR, MMLU-Pro), whose scores we take as published.

3 SETTING AND DEFINITIONS

Leaderboard and rules. A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of M models on n tasks, taken from a fixed public snapshot. An aggregation rule f induces a ranking; we study the *score rules* arithmetic mean, median (mean of the middle pair for even n), minimum, and shifted geometric and harmonic means (computed on score$+1$ so that zeros are admissible, then shifted back), and the *ordinal rules* Borda (summed per-task ranks), plurality (per-task first places, ties split equally), and, as a requirement rather than a ranking, the *Condorcet requirement* (beat every rival in a strict task majority, task ties counting half). Because a Condorcet-consistent rule can also elect a model that is not a Condorcet winner, the Condorcet cost is an upper bound for such rules, not their cost. Trimmed-mean and power-mean variants are sensitivity checks.

Investment. Fix a target model with score vector $x \in \mathbb{R}^n$ and hold all rival rows fixed. Targeted investment raises the target's score on task t to $y_t \in [x_t, c_t]$, where c_t is a *cap* (the effective cap is $\max(x_t, \text{scenario cap})$, with scenario caps clipped at 100, so a target that already exceeds a scenario cap on a task keeps its original score there instead of facing an empty box); this stands for training on data related to sub-dataset t. The target is *first* if $f(y) \geq \max_r f(x_r) + \varepsilon$ for score rules ($\varepsilon = 0.001$ points; ordinal rules use an analogous finite convention: per-task levels are the original score, a tie with a rival score, or one margin step above it, taking only rival scores at or above the target's original score on that task), and a target that is already uniquely first has cost zero. We measure investment in three ways, each optimized separately: $K = \|y - x\|_0$ (tasks touched), $L_1 = \sum_t (y_t - x_t)$ (score points) and $L_\infty = \max_t (y_t - x_t)$ (the largest single-task jump, which is the budget of a uniform "general improvement"). Their minimizers need not coincide. Infeasible cases (no vector within the caps makes the target first) have infinite cost and are reported as a feasible fraction, never dropped.

Caps are the modelling assumption. Public scores identify no achievable training gain. We therefore treat the cap as a scenario and show every result with it: `ceil` ($c_t = 100$; what is needed regardless of realism), `front0/2` (the best rival score on t plus 0 or 2 points), `gain5/10/20` (at most $+5/10/20$ points per task) and, for the Open LLM Leaderboard only, `emp95/99`, per-task quantiles of observed fine-tune-minus-base score differences. The latter are descriptive scales, confounded by selection, architecture and training, and are *not* estimates of achievable gains.

Exact results. The following hold for the model above (proofs in Appendix C; `analysis/` checks each against the solvers). Throughout, a target that is already uniquely first costs 0 by convention, even if it wins by less than ε; the formulas below are for the other targets. (i) *Mean:* with gap $g = \tau + \varepsilon - \bar{x}$ to the best rival mean τ, minimum $L_1 = n\,g$ whenever total headroom suffices, and minimum K is the shortest descending-headroom prefix reaching $n\,g$. (ii) *Minimum:* with $b =$ best rival minimum $+\varepsilon$, feasibility requires every cap $\geq b$; then $K = \#\{t : x_t < b\}$ and $L_1 = \sum_t \max(0, b - x_t)$. (iii) *Odd median* ($n = 2h - 1$): at least h final coordinates must reach b; L_1 is the sum of the h cheapest *cap-feasible* lifts (infeasible if fewer than h coordinates can reach b). (iv) *Cap threshold:* for any rule whose set of first-placing plans in the ceiling box is closed (all rules studied here), a gain-h scenario is feasible iff the `ceil` value of L_∞ is $\leq h$. (v) *No universal order:* the costs of the mean, median and minimum rules can reverse (Table 1), so no universal ordering of these rules and no blanket statement such as "the median needs half the tasks" holds; we do not prove reversals for every other rule studied. (vi) *Direction of the bounds:* the score-rule minima are exact minima in this fixed-rival box model, hence lower bounds on the score-space cost of any intervention feasible in the same box, not on the cost of an actual training plan. The ordinal minima are exact only on the declared finite set of per-task levels (unchanged, tie a rival, pass a rival by one margin step), so they bound the infimum over arbitrary real scores from above, and infeasibility on that set is conditional on it. The Condorcet requirement gives a different, sufficient-condition upper bound for a Condorcet-consistent rule.

Table 1: Constructed (not measured) examples with $n = 5$ tasks, one rival row and margin $\varepsilon = 10^{-3}$: exact minimum K and L_1 for the target to be first. The cheapest rule changes between examples; in the last one the target already wins under the median (cost 0). Source: `analysis/results/theory_examples.json`.

				mean		median		min	
example	target	rival	cap	K	L_1	K	L_1	K	L_1
concentrated headroom	$(10, 0, 0, 0, 0)$	$(3, 3, 3, 3, 3)$	100	1	5.005	2	6.002	4	12.004
concentrated rival	$(0, 0, 0, 0, 0)$	$(9, 9, 1, 1, 1)$	5	5	21.005	3	3.003	5	5.005
mostly above bottleneck	$(4, 4, 4, 4, 0)$	$(9, 9, 1, 1, 1)$	$(5.1, 5.1, 5.1, 5.1, 1.1)$	5	5.005	0	0	1	1.001

4 Data and methods

Frames. *HELM MMLU* (release v1.13.0, dated 2025-01-10): 79 models, each with all 57 per-subject exact-match accuracies (the 57 subject tables, not the aggregate). *Open LLM Leaderboard* `contents` (commit `9c09a7c`…, 2025-03-20): 4,576 rows; after removing repeated model names (keeping the largest reported average, 79 repeated names) and one flagged model, 4,496 models on six normalised tasks. A fixed top-500 frame (selected once by the normalised mean) is used for ordinal rules, a no-merge frame (3,789 models) and the raw (unnormalised) scale as sensitivity checks. These are censuses of two snapshots, not samples of models, and they are never pooled. Source licences are undeclared for the scores (Appendix D); checksums and retrieval dates are recorded in the supplement.

Targets and comparisons. Every model in a frame is a target. For rule comparisons we pair targets across rules; since the mean's unique leader has cost zero under the mean by convention, it is excluded when conditioning on non-leading targets, and targets already first under another rule keep cost zero. We report costs against the target's initial rank under the mean, as well as over all targets.

Solvers. Score rules use closed forms where available (mean, minimum, odd median), water-filling for concave power means, and exact enumeration of threshold pairs for the even median. Trimmed means (supplementary; L_1 and L_∞ only) are continuous score rules, solved by a mixed-integer program with sorting assignments (L_1) and by bisection (L_∞); a rerun with an explicit zero solver gap reproduced the supplementary output exactly (500 rows, 100 of them trimmed means). Only the ordinal rules (Borda, Condorcet, plurality) are solved exactly over the finite per-task level set "unchanged / tie a rival / pass a rival by one margin step" (only rival scores at or above the target's original score on the task are candidate levels), with zero requested optimality gap. A solver limit or numerical failure raises an error and is never recorded as infeasibility. The solvers are tested against exhaustive search and independent LP/convex oracles on small random instances, and a result checker verifies, on the full outputs, the mean-gap identity, monotonicity in the cap, agreement of feasibility across K, L_1, L_∞, and the cap-threshold equivalence. For Borda a dynamic program is used, checked by a witness test with a full mixed-integer fallback; on the actual Open LLM Leaderboard frame it agrees with an independently built full-rival mixed-integer program in all 24 checked cases (six target ranks, ceiling and 10-point cap, L_1 and K), and across the 4,948 Borda objective instances of the HELM and top-500 mains none required the fallback (an audit re-run, not telemetry of the original runs).

Uncertainty. Cost summaries over targets describe the fixed frame. To probe sensitivity we (a) resample tasks with replacement (100 shared draws; original-mean ranks 2–11 followed under every rule), (b) perturb scores with Gaussian noise scaled by plug-in binomial standard errors, clipped to $[0, 100]$, with per-task shock correlation 0 and 0.8, and (c) vary the number of HELM subjects ($n \in \{3, 6, 12, 24, 48, 57\}$, 100 subsets). The resulting percentile ranges are sensitivity intervals under stated assumptions, not population confidence intervals.

Margin and rival-set checks. On the 20 targets of original mean rank 2–21 per frame we change ε from 0.001 to 0.1; on ranks 2–11 we recompute all rules with the mean's leader removed and under ten random half-frames of the rivals (the target is always kept).

Table 2: Minimum investment to become uniquely first, all non-leading targets of each fixed census: median [Q1, Q3] (inverted-CDF convention: lower middle value for even N) of tasks touched K and total score points L_1 under the ceiling cap (scores may rise to 100) and under a 10-point per-task cap. feas.: targets that can reach first place; †: median over feasible targets only (a different target set for each rule). Condorcet winner is a requirement, an upper bound for Condorcet-consistent rules. Ordinal rules: HELM and the top-500 frame only (separate block; the frames are not pooled). Source: `analysis/results/summary/main_descriptive.csv`.

	cap ceil			cap gain10		
rule	feas.	K	L_1	feas.	K	L_1
OLLB (6 tasks), N = 4495 non-leading targets						
mean	4495/4495	2 [2, 3]	181 [137, 229]	113/4495	6†	50.9†
geometric mean	4495/4495	3 [2, 4]	153 [113, 201]	142/4495	5†	48.2†
harmonic mean	4495/4495	3 [2, 4]	125 [90.0, 171]	264/4495	5†	39.5†
median	4495/4495	3 [3, 3]	106 [78.5, 146]	203/4495	3†	18.3†
min	4495/4495	3 [3, 5]	43.2 [29.5, 67.8]	446/4495	2†	10.4†
HELM MMLU (57 subjects), N = 78 non-leading targets						
mean	78/78	16 [10, 23]	661 [440, 1057]	29/78	37†	360†
geometric mean	78/78	16 [10, 23]	633 [421, 1028]	24/78	30†	292†
harmonic mean	78/78	16 [10, 23]	609 [404, 1001]	20/78	28†	270†
median	78/78	21 [14, 27]	111 [37.7, 257]	40/78	14†	37.7†
min	78/78	11 [7, 18]	97.5 [45.9, 242]	14/78	3†	7.6†
Borda	78/78	46 [38, 52]	500 [291, 893]	25/78	40†	185†
Condorcet winner	78/78	29 [28, 29]	213 [108, 354]	38/78	28†	106†
plurality	78/78	9 [9, 9]	46.7 [25.3, 79.8]	57/78	9†	31.0†
OLLB top-500 frame (6 tasks; ordinal rules), N = 499 non-leading targets						
mean	499/499	1 [1, 1]	71.1 [61.6, 82.0]	113/499	6†	50.9†
geometric mean	499/499	1 [1, 2]	48.5 [42.1, 56.4]	142/499	5†	48.2†
harmonic mean	499/499	2 [1, 2]	31.9 [27.1, 37.2]	261/499	5†	39.4†
median	499/499	2 [2, 2]	26.5 [20.5, 34.4]	200/499	3†	18.3†
min	499/499	2 [2, 2]	12.3 [8.4, 19.4]	348/499	2†	9.6†
Borda	499/499	4 [3, 4]	36.4 [27.1, 47.0]	325/499	4†	30.6†
Condorcet winner	499/499	4 [4, 4]	47.0 [39.3, 54.8]	60/499	4†	23.3†
plurality	499/499	2 [2, 2]	26.3 [23.0, 28.6]	53/499	2†	13.1†

Controls. For every feasible non-leading target we compare the optimal K and L_1 with four policies: raising tasks to their caps until the target is first in order of lowest original score, of largest headroom, or in random order (median over 200 orders, seed 7), and the uniform improvement that attains the optimal L_∞; the ratio ranges come from 1000 target resamplings shared by all policies. The earlier 20-target random-order and uniform comparison (original mean ranks 2–21) is kept in the appendix.

5 Experiments

We report fixed-frame census summaries: medians over all non-leading targets (the mean's unique leader is excluded and costs zero under it), with the interquartile range in brackets, and, whenever a cap is binding, the feasible fraction next to every cost. Raw K and L_1 are not comparable between the 6-task and 57-task frames. Medians and quartiles over targets use the inverted-CDF definition (for an even number of targets, such as the 78 HELM targets, the lower of the two middle values); the usual midpoint median is given in the text where it differs (for example 11.5 against 11 for the minimum rule's K on HELM) and for every cell in Appendix Tables 10–13 (with resampling ranges for the medians and feasible counts). A range in brackets after a count or share is the 2.5–97.5 percentile range over 1000 resamplings of the targets of the fixed census (seed 23): a sensitivity range for that census, not a confidence interval for a population of models; Wilson intervals, a binomial reference, are in the same file. Costs for every target, cap and rule are in `analysis/results/*.csv`; the table and figure to script map is in `figtab/README.md`.

5.1 How much is needed, and how it varies with the rule

Unrestricted headroom. With every score allowed to rise to 100 (Table 2, left), a typical non-leading Open LLM Leaderboard model needs to raise 2 of its 6 tasks and 181 points in total under the mean; the median is 3 tasks and 106 points, the minimum 3 tasks and 43 points. On HELM the mean needs 16 of 57 subjects and 661 points (midpoint median 696), the minimum 11 subjects (11.5) and 98 points, plurality 9 subjects and 47 points. Per task, the mean's typical HELM target needs 11.6 points on 28% of the subjects, against 30.1 points on 33% of the tasks on the Open LLM Leaderboard; raw units are not commensurable between the frames. Because the minimum rule only requires lifting a model's weakest tasks to the rival's weakest score, whereas the mean requires closing n times the gap in the average, the *points* needed are lower under every other rule we compare than under the mean for at least 98.7% of targets (paired on the same target, Appendix Fig. 6), on both frames, in raw points; in units of each task's standard deviation the mean no longer has the largest median L_1 among the score rules (geometric mean above it in all three frames, Appendix Table 9). The number of *tasks* tells a different story and depends on the frame: on HELM the minimum needs fewer tasks than the mean for 91% of targets and plurality for 81%, whereas the median needs more for 82%, Borda for all targets and the Condorcet requirement for 87%; on the Open LLM Leaderboard the minimum needs more tasks than the mean for 81% of targets (equal for 19%) and the median needs the same number for 53%. These comparisons mix the rule's response to increases with the different gap each rule induces; the constructed examples (Table 1) show that the sign can flip, so none of this is a universal ordering. Cost tends to grow with the target's rank for every rule (Appendix Fig. 22).

The headroom assumption decides whether first place is reachable. With at most $+10$ points per task, most targets cannot reach first place under any rule (Table 2, right; Appendix Fig. 5a): 113 of 4,495 Open LLM Leaderboard targets (2.5% [2.0, 3.0]) under the mean, 446 (9.9% [9.0, 10.8]) under the minimum; on HELM 29 of 78 (37% [26, 49]) under the mean, 14 (18% [10, 27]) under the minimum, but 57 (73% [63, 83]) under plurality. The $+10$ cap is a scenario with no data-based justification; at $+5$ and $+20$ the mean reaches first place for 13 (0.3%) and 830 (18%) Open LLM Leaderboard targets and for 11 (14%) and 56 (72%) HELM targets (Appendix Table 16), so the feasibility percentages are properties of the cap and not of the leaderboards. Resampling the targets of each census with replacement (1000 draws, seed 23; analysis/results/paired_target_bootstrap.csv; a sensitivity range for the selected census, not a population confidence interval; Appendix Fig. 16) gives for the difference in feasible fraction at the 10-point cap, minimum minus mean, -0.19 [-0.29, -0.09] on HELM, $+0.07$ [0.07, 0.08] on the 4,496-model Open LLM Leaderboard and $+0.47$ [0.41, 0.52] on the top-500 frame;. At $+10$ the minimum is thus the most reachable score rule on the Open LLM Leaderboard and the least reachable on HELM (Section 5.3 asks how far that is a property of the frames). By the threshold equivalence of Section 3, the feasible fraction under a gain-h cap is exactly the fraction of targets whose ceiling-cap L_∞ is at most h, which is why L_∞ is reported as a feasibility diagnostic. The empirical gain scales (`emp95/99`) are descriptive only.

5.2 Does targeting matter?

A factor of 13–15 in tasks for the minimum rule against random order on 20 near-top HELM targets concerns a naive baseline on few targets, so we compare the optimum with greedy orders over *all* feasible targets of each frame (analysis/results/final_controls_summary.csv; 110,876 target-rule-cap records; ranges from 1000 target resamplings shared by all policies). Raising tasks in order of lowest score first attains the optimal K for every target at the ceiling cap (Figure 1), for all five score rules and all three frames (15 panels), and at both caps it is optimal for the minimum rule, as highest-headroom-first is for the mean (Propositions 1–2). Targeting thus beats random order, whose median task count over the 78 HELM targets at the ceiling is 1.76 [1.67, 1.92] (mean), 1.14 [1.08, 1.21] (median) and 4.23 [3.44, 6.00] (minimum) times the optimum, but shows no advantage in K over a greedy rule that needs no optimiser. Under the 10-point cap the median rule is where the optimiser matters: lowest-score-first needs 2.6 [2.4, 2.8] times the optimal K on HELM (1.7 on the Open LLM Leaderboard) and matches it for none of 40 HELM targets. In points, cap-prefix greedy policies have median L_1 7.7 times (median rule) and 4.8 times (minimum) the optimum on HELM at the ceiling, against 1.0 under the mean (Appendix Fig. 23). These describe the named policies, not training costs.

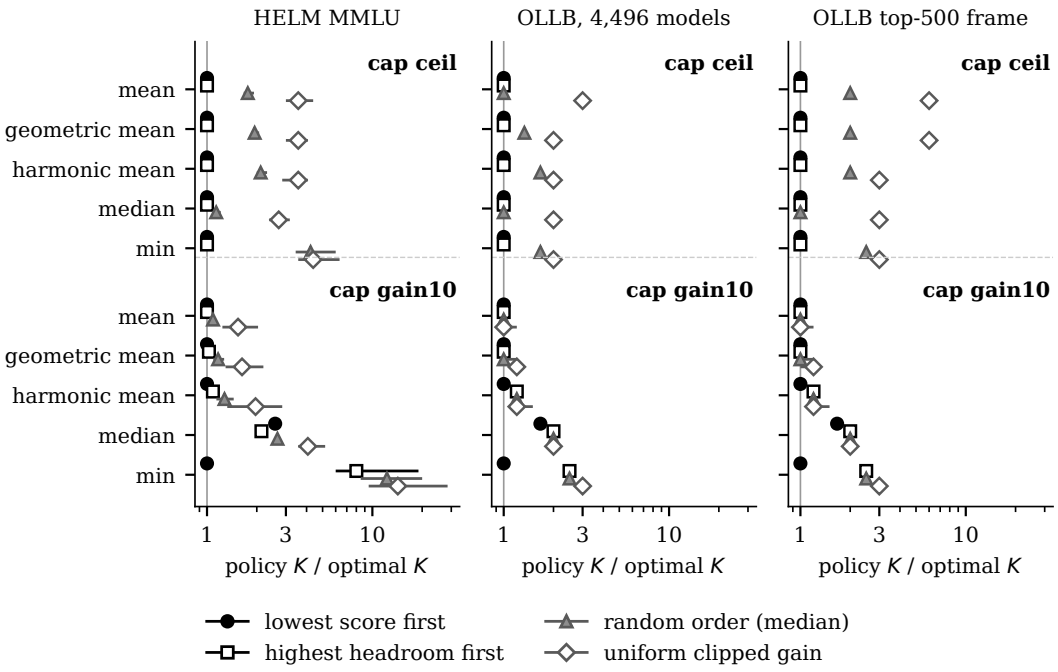


Figure 1: Tasks needed by four policies relative to the optimal K (log axis; median over all feasible targets, bars: 2.5–97.5 percentile range over 1000 target resamplings shared across policies; numbers of feasible targets are in Appendix Table 12). Lowest-score-first coincides with the optimum at the ceiling for every score rule and frame. Source: `figures/fig11m_policy_controls_main.provenance.json`.

Table 3: Which comparisons of the minimum rule (min) with the mean survive the checks. S: stable; R: reversed; —: not changed by the check. Full frames: HELM, Open LLM Leaderboard (OLLB), top-500 (T500). Matched frames: random 79-model OLLB subsets, mean-top-79 frame, random six-subject HELM subsets (HELM-6); median or majority sign. SD units: costs per task standard deviation. IQR cap: cap at each task's interquartile range. Records which orderings change, not why. Sources: Table 2; Appendix Tables 5, 6, 9, 7, 8.

comparison	HELM / OLLB / T500	random OLLB-79 / top-79 / HELM-6	SD units	IQR cap (HELM / OLLB / T500)
Fewer tasks K at ceiling	min / mean / mean	S / S / R	—	—
Lower L_1 at ceiling	min / min / min	S / S / S	S	—
Mean costliest score rule in L_1	yes / yes / yes	S / S / S	R	—
More reachable at $+10$	mean / min / min	S (84) / R / S (89)	—	S / R / S

5.3 What the reversal depends on

At the 10-point cap the minimum rule is more reachable than the mean on the Open LLM Leaderboard and less reachable on HELM. Matching one dimension and the cap and unit change this picture (Figure 2; 100 random frames per row, seed 31; ranges across selected frames are sensitivity ranges, not confidence intervals). Among 100 random 79-model subsets of the Open LLM Leaderboard (the number of rivals matched to HELM) the minimum is more reachable than the mean in 84 draws, less in 12 and equal in 4 (median paired difference $+0.08$ [-0.05, $+0.31$]); in the fixed top-79 frame selected by the mean it is less reachable (59 of 78 against 78 of 78, -0.24). Among 100 random six-subject subsets of HELM (the task count matched to the Open LLM Leaderboard) the minimum is less reachable in 89 draws and more in 9 (-0.18 [-0.37, $+0.23$]). So each sign survives matching in most random draws, with exceptions, and the mean-selected frame reverses it; random subsets keep the task correlation of their parent frame, mean-selected frames have almost none (0.02 and 0.08; Appendix Table 5), but the frames differ in several respects at once and none

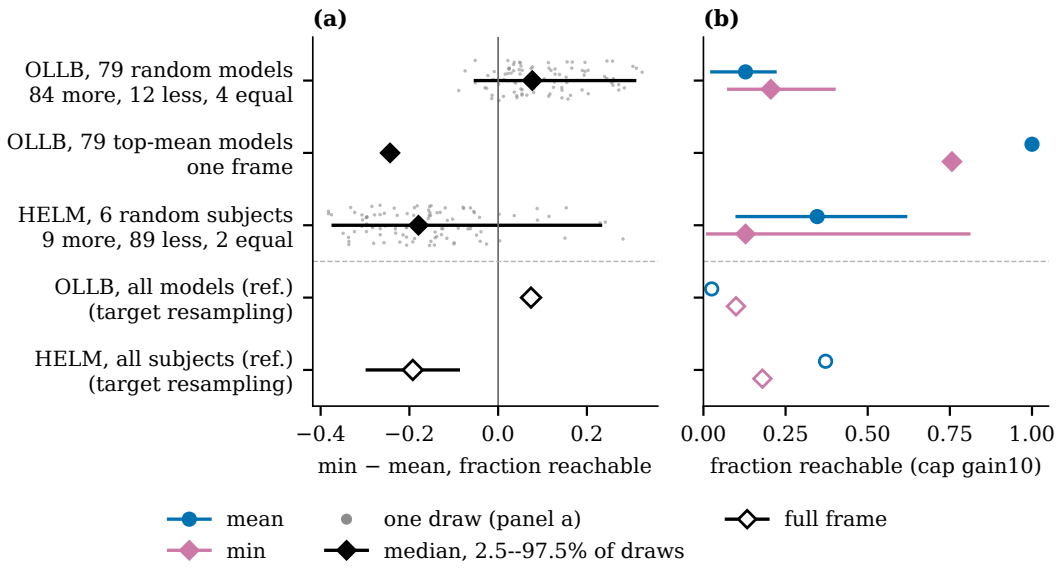


Figure 2: Minimum minus mean in the fraction of targets reachable at the 10-point cap, over frames matched in rivals (random 79-model subsets of the Open LLM Leaderboard, and the fixed mean-top-79 frame) or tasks (random six-subject subsets of HELM): (a) paired difference per frame, black: median and 2.5–97.5% range over 100 frames (seed 31); counts: frames where the minimum is more, less or equally reachable; (b) the fractions; open: full frames. Ranges over selected frames are not confidence intervals. Source: `figures/fig13_frames.provenance.json`; Appendix Tables 5, 8.

of this identifies a cause. The cap matters as much (Appendix Fig. 3, Tables 6, 4): under a cap equal to each task's interquartile range of scores (a descriptive scale, not an achievable gain) the full Open LLM Leaderboard reverses to the mean 582 against the minimum 283 of 4,495 targets, whereas the minimum leads on the top-500 frame (129 against 52 of 499) and the mean on HELM (39 against 20 of 78). The reachability ranking of the rules therefore depends on frame, cap and unit, and the $+10$ reversal is not a property of the leaderboards (Table 3 lists which comparisons survive).

5.4 Sensitivity

Resampling the tasks (100 draws, ten near-top targets per frame, cap $+10$) changes comparisons substantially: on HELM, whether the minimum rule can reach first place varies strongly across draws (feasible in 0.11 to 1.0 of the draws depending on the target, against at least 0.88 for every other rule). Score noise and the HELM task-count experiment give percentile ranges under stated assumptions, not confidence intervals (Appendix B). On 20 near-top targets per frame, the margin ε and the rival set also change task counts (Appendix Tables 21, 22).

Ordinal rules on the top-500 Open LLM Leaderboard frame. The ordinal rules are computed on a fixed top-500 frame (selected once by the normalised mean; 499 non-leading targets), kept separate from the 4,496-model census and from HELM (Table 2, separate block; Appendix Table 20). At the ceiling the median target needs $K = 4$ (Borda), 2 (plurality) and 1 (mean) tasks; at the 10-point cap the reachable targets are Borda 325, plurality 53, mean 113, minimum 348 and the Condorcet requirement 60 of 499, a ranking unlike HELM's.

Processing choices in the Open LLM Leaderboard frame. The main frame keeps the largest reported average for each of 79 repeated names (74 of the 79 have a lower score on some task than the lowest-average duplicate) and drops one flagged model. Keeping the lowest average, including the flagged model, or facing only non-merged, unflagged rivals leaves the 10-point feasibility ordering of the score rules and the ceiling-cap L_1 order unchanged (reachable for the mean: 113 of 4,495

in the main frame, 111, 113, and 81 of 3,788 without merged models; Appendix Table 15). The *raw* score scale changes the 10-point ordering (reachable: harmonic 459, geometric 382, mean 296, minimum 224, median 21), as $+10$ raw points is a different intervention from $+10$ normalised points. The mean-selected top-500 frame lowers the mean's cost by construction ($L_1 = 71$ against 181).

6 Discussion and limitations

What the results say. On the two public frames, the investment needed to move a model to first place is governed jointly by the aggregation rule, the cap on per-task improvement and where the target stands. For a non-leading target the mean's cost is proportional to the gap ($L_1 = n(\tau + \varepsilon - \bar{x})$ when headroom suffices, Section 3); rules that respond to the lowest or middle scores can be far cheaper in points, but a binding cap can make first place unreachable under them. The constructed examples show that the ranking of the mean, median and minimum rules can reverse, and the HELM census contradicts a blanket claim that the minimum must need the most tasks or points. Targeting matters less than an earlier version suggested (Section 5.2): a lowest-score-first greedy order attains the optimal task count at the ceiling for every score rule, so the optimiser adds value mainly in points and under a binding cap for the median rule.

Modelling assumptions. (i) The developer knows every score and all rivals are fixed; the minima hold only within this box model (fixed rivals, score-space cost),; public scores give no points-to-effort conversion. (ii) Tasks are controlled independently, although real training changes many tasks together. (iii) The caps are scenarios, not estimates of achievable gains. (iv) L_1 counts every point as equally costly. (v) "First" means strictly ahead by 0.001 points; changing ε to 0.1 changed some task counts and point costs. (vi) The ordinal costs are exact only on the declared finite level set and upper-bound the real-valued infimum; infeasibility there is conditional on it; the Condorcet requirement is a sufficient-condition upper bound. (vii) The rival set is a convention (Section 5).

Data limits. Both frames are fixed censuses, not samples (HELM: 79 models, 57 correlated subjects; Open LLM Leaderboard: six tasks, merged and fine-tuned models): all ranges are sensitivity ranges, not confidence intervals.

Why the rules reverse between frames. The frames differ at once in task count, rivals, task correlation, score scale and population; matching, cap and unit change the picture (Section 5.3). We name no single cause and claim nothing for other leaderboards.

Reproduction. The original 16-file study and the review and final additions were each recomputed in isolated clean exports and agreed (maximum finite difference 5.8×10^{-9} in the original, zero in the additions; Appendix E). That pass used the bundled pinned processed matrices; re-downloading the raw sources was not repeated.

Ethics statement

All data are public scores; no model is trained and no human subjects are involved. The study could inform strategic gaming of leaderboards, but it is intended to help benchmark designers judge how aggregation choices shape incentives, and it makes no claim about any developer's practices.

Reproducibility statement

Code, pinned processed tables with SHA-256 checksums, seeds, commands, measured run times (Appendix E), a table/figure-to-script map and the status ledger are in the supplement (`analysis/`, `README.md`); data sources and licences are in Appendix D.

References

Felix Brandt, Vincent Conitzer, Ulle Endriss, Jérôme Lang, and Ariel D. Procaccia (eds.). *Handbook of Computational Social Choice*. Cambridge University Press, 2016. doi: 10.1017/CBO9781107446984.

Yatong Chen, Guanhua Zhang, and Moritz Hardt. Leaderboard incentives: Model rankings under strategic post-training. *arXiv preprint arXiv:2603.08371*, 2026.

Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. *arXiv preprint arXiv:2107.07002*, 2021.

Piotr Faliszewski, Edith Hemaspaandra, and Lane A. Hemaspaandra. The complexity of bribery in elections. In *Proceedings of the 21st AAAI Conference on Artificial Intelligence*, 2006.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. *arXiv preprint arXiv:2009.03300*, 2020.

Marc Lanctot, Kate Larson, Yoram Bachrach, Luke Marris, Zun Li, Avishkar Bhoopchand, Thomas Anthony, Brian Tanner, and Anna Koop. Evaluating agents using social choice theory. *arXiv preprint arXiv:2312.03121*, 2023.

Percy Liang, Rishi Bommasani, Tony Lee, Dimitris Tsipras, Dilara Soylu, Michihiro Yasunaga, Yian Zhang, Deepak Narayanan, Yuhuai Wu, Ananya Kumar, et al. Holistic evaluation of language models. *arXiv preprint arXiv:2211.09110*, 2022.

Shivalika Singh, Yiyang Nan, Alex Wang, Daniel D'Souza, Sayash Kapoor, Ahmet Üstün, Sanmi Koyejo, Yuntian Deng, Shayne Longpre, Noah A. Smith, Beyza Ermis, Marzieh Fadaee, and Sara Hooker. The leaderboard illusion. *arXiv preprint arXiv:2504.20879*, 2025.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. Challenging BIG-Bench tasks and whether chain-of-thought can solve them. *arXiv preprint arXiv:2210.09261*, 2022.

Shuo Yang, Wei-Lin Chiang, Lianmin Zheng, Joseph E. Gonzalez, and Ion Stoica. Rethinking benchmark and contamination for language models with rephrased samples. *arXiv preprint arXiv:2311.04850*, 2023.

Guanhua Zhang and Moritz Hardt. Inherent trade-offs between diversity and stability in multi-task benchmarks. *arXiv preprint arXiv:2405.01719*, 2024.

Kun Zhou, Yutao Zhu, Zhipeng Chen, Wentong Chen, Wayne Xin Zhao, Xu Chen, Yankai Lin, Ji-Rong Wen, and Jiawei Han. Don't make your LLM an evaluation benchmark cheater. *arXiv preprint arXiv:2311.01964*, 2023.

A AI USE STATEMENT

This study, including its research design, mathematical statements and proofs, experiment design, implementation, data cleaning, execution of experiments, interpretation of results, figures, literature search and the writing of this manuscript, was carried out by AI language-model agents working as separate roles (planning, analysis, presentation, venue, report writing and independent review) in an automated research organisation. No human wrote the content. All hypotheses, the investment model, the aggregation rules and the claims were proposed by the agents. Verification consisted of unit tests of the solvers against brute-force search and independent LP/convex oracles, an automated invariant checker on the full result files, independent review by a separate agent role, and, for the completed runs, recomputation of summaries and figures from the stored result files by the scripts in the supplement, and a clean recomputation of every result file from the pinned processed matrices (Appendix E). Only references whose abstract or text an agent actually consulted are cited. Agents may nevertheless have made errors that these checks did not catch.

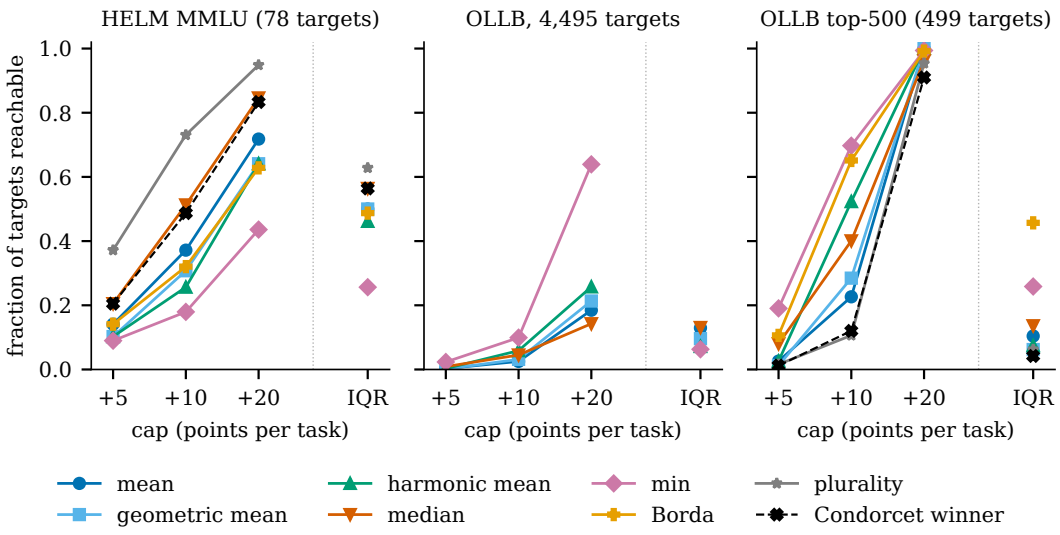


Figure 3: Which targets can become strictly first depends on the cap as well as on the rule. Fraction of the mean-non-leading targets of each fixed frame that can be made first, at the smallest number of tasks touched, when each task can rise by at most 5, 10 or 20 points, or by the frame's own task interquartile range (IQR; effective cap never below the target's score), at most 100. The full OLLB frame has the five score rules; Borda, plurality and the Condorcet requirement (dashed, a sufficient condition) are on HELM and the top-500 frame only. Under +10 the minimum rule is the most reachable score rule on full OLLB (446/4,495 vs 113 for the mean); under the IQR cap the sign reverses (283 vs 582). On HELM the minimum is the least reachable score rule at +10 and under IQR. Counts of fixed censuses; the IQR cap is a descriptive data-derived scale, not an achievable training gain. At the ceiling every target is reachable. Source: `figures/fig14_caps.provenance.json`.

Table 4: Targets that can become first (of N mean-non-leading targets) under gains of at most 5, 10 or 20 points per task or the task-IQR cap. Source: `analysis/results/final_caps_feasibility.csv`.

frame (N)	rule	+5	+10	+20	IQR
HELM (78)	mean	11	29	56	39
	geometric mean	8	24	50	39
	harmonic mean	8	20	50	36
	median	16	40	66	44
	min	7	14	34	20
	Borda	11	25	49	38
	plurality	29	57	74	49
	Condorcet winner (req.)	16	38	65	44
OLLB (4495)	mean	13	113	830	582
	geometric mean	6	142	959	432
	harmonic mean	13	264	1157	322
	median	40	203	641	588
	min	105	446	2873	283
OLLB top-500 (499)	mean	13	113	499	52
	geometric mean	6	142	499	31
	harmonic mean	13	261	499	37
	median	40	200	480	68
	min	95	348	496	129
	Borda	53	325	495	228
	plurality	8	53	476	31
	Condorcet winner (req.)	6	60	454	21

Table 5: Frame isolation at cap gain10 (at most $+10$ points per task): fraction of the mean-non-leading targets that can be made strictly first by the mean rule and by the minimum rule, and their paired difference (min $-$ mean). Fixed frames (HELM, full OLLB, OLLB top-500) show the fraction over all their targets; * the bracket is the 2.5–97.5% target-resampling range (R1000, seed 23) of the paired difference. Matched-frame families keep the 79 models and 78 targets: 100 random 79-model OLLB subsets, 100 random 6-subject HELM subsets (all 79 models), and one fixed OLLB frame of the 79 highest-mean models; entries are median [2.5, 97.5 percentiles] across frames, selected-frame sensitivity ranges, not confidence intervals. $+/-/=$: frames in which the minimum is more, less, equally reachable than the mean. Task corr.: mean pairwise correlation of task scores; gap: mean (best-rival threshold under the minimum rule minus under the mean rule) in score points. Task correlation and gap were supplied only for the matched-frame families; the frames differ in several respects at once, so no single cause is identified. At the ceiling every target is reachable and the difference is 0 in all frames.

frame	targets	frames	mean	min	min − mean	$+/-/=$	task corr.	gap
HELM, all 57 subjects	78	–	0.37	0.18	−0.19 [−0.29, −0.09]*	–	0.79	−28
HELM, 6 random subjects	78	100	0.35 [0.10, 0.62]	0.13 [0.01, 0.81]	−0.18 [−0.37, +0.23]	9/89/2	0.80 [0.66, 0.88]	−8 [−27, −3]
OLLB, all models	4495	–	0.03	0.10	+0.07 [+0.07, +0.08]*	–	0.67	−29
OLLB, 79 random models	78	100	0.13 [0.03, 0.22]	0.21 [0.08, 0.40]	+0.08 [−0.05, +0.31]	84/12/4	0.67 [0.58, 0.74]	−26 [−31, −21]
OLLB, 79 top-mean models	78	1	1.00	0.76	−0.24	–	0.02	−29
OLLB top-500	499	–	0.23	0.70	+0.47 [+0.41, +0.52]*	–	0.08	−29

Table 6: Cap dependence: number of the mean-non-leading targets that can be made strictly first by the mean, minimum and median rules (mean/min/median) when each task can rise by at most 5, 10 or 20 points or by the frame's task interquartile range (IQR). The superscript is the sign of min minus mean ($+$ minimum reaches more targets, $-$ fewer). The full-OLLB advantage of the minimum at $+5$, $+10$, $+20$ reverses under the IQR cap (582 mean vs 283 minimum of 4,495); On top-500 the minimum is ahead at +5, +10 and IQR and behind at +20 (496 vs 499); on HELM it is behind at every cap. Counts of fixed censuses; the IQR cap is a descriptive data-derived scale, not an achievable training gain.

frame	targets	$+5$	$+10$	$+20$	IQR
HELM	78	$11/7/16^{-}$	$29/14/40^{-}$	$56/34/66^{-}$	$39/20/44^{-}$
OLLB, all models	4495	$13/105/40^{+}$	$113/446/203^{+}$	$830/2873/641^{+}$	$582/283/588^{-}$
OLLB top-500	499	$13/95/40^{+}$	$113/348/200^{+}$	$499/496/480^{-}$	$52/129/68^{+}$

Table 7: Matched-frame families, cap ceil: for each rule or requirement, the fraction of the 78 mean-non-leading targets that can be made first, and the median K and L_1 over all targets, as the median [2.5, 97.5 percentiles] across the frames of the family (number of frames in brackets after the family). Within a frame the median counts an infeasible target as $+\infty$, so it is ∞ when more than half the targets are infeasible; † median over the feasible targets only (inverted-CDF convention). Condorcet is a requirement (an upper bound). Selected-frame sensitivity ranges, not confidence intervals; the fixed top-79 frame is a single frame. Source: `analysis/results/final_frame_intervals.csv`.

family	rule	feasible	K	L_1
OLLB, 79 random models (100)	mean	1.00	2	138 [109, 178]
	geometric mean	1.00	2 [2, 3]	105 [81.8, 150]
	harmonic mean	1.00	3 [2, 3]	81.8 [55.6, 121]
	median	1.00	3	87.8 [73.4, 116]
	min	1.00	3	30.8 [21.8, 41.9]
	Borda	1.00	5 [4, 6]	109 [83.1, 142]
	Condorcet winner	1.00	4	76.7 [63.4, 103]
	plurality	1.00	2 [2, 3]	44.2 [23.2, 68.6]
OLLB, 79 top-mean models (1)	mean	1.00	1	43.4
	geometric mean	1.00	1	29.4
	harmonic mean	1.00	1	21.5
	median	1.00	2	14.2
	min	1.00	2	11.1
	Borda	1.00	2	16.7
	Condorcet winner	1.00	3	23.4
	plurality	1.00	2	13.3
HELM, 6 random subjects (100)	mean	1.00	2 [2, 3]	73.0 [40.7, 121]
	geometric mean	1.00	3 [2, 3]	72.5 [36.4, 120]
	harmonic mean	1.00	3 [1, 3]	71.2 [33.8, 119]
	median	1.00	3 [2, 3]	28.9 [15.2, 54.2]
	min	1.00	3 [1, 6]	43.3 [4.8, 92.0]
	Borda	1.00	6 [5, 6]	58.6 [30.2, 101]
	Condorcet winner	1.00	4	35.1 [19.5, 60.4]
	plurality	1.00	2 [2, 3]	13.6 [6.3, 32.8]

Table 8: Matched-frame families, cap gain10: for each rule or requirement, the fraction of the 78 mean-non-leading targets that can be made first, and the median K and L_1 over all targets, as the median [2.5, 97.5 percentiles] across the frames of the family (number of frames in brackets after the family). Within a frame the median counts an infeasible target as $+\infty$, so it is ∞ when more than half the targets are infeasible; † median over the feasible targets only (inverted-CDF convention). Condorcet is a requirement (an upper bound). Selected-frame sensitivity ranges, not confidence intervals; the fixed top-79 frame is a single frame. Source: `analysis/results/final_frame_intervals.csv`.

family	rule	feasible	K	L_1
OLLB, 79 random models (100)	mean	0.13 [0.03, 0.22]	∞; 4 [2, 6]†	∞; 36.5 [14.6, 53.6]†
	geometric mean	0.15 [0.04, 0.23]	∞; 3 [1, 5]†	∞; 20.5 [8.7, 48.7]†
	harmonic mean	0.15 [0.05, 0.24]	∞; 2 [1, 5]†	∞; 17.2 [7.6, 40.0]†
	median	0.06 [0.01, 0.14]	∞; 2 [1, 3]†	∞; 11.2 [1.0, 23.2]†
	min	0.21 [0.08, 0.40]	∞; 2†	∞; 10.4 [4.8, 15.1]†
	Borda	0.13 [0.03, 0.22]	∞; 3 [1, 5]†	∞; 16.9 [2.1, 31.1]†
	Condorcet winner	0.10 [0.00, 0.17]	∞; 3 [2, 4]†	∞; 12.4 [5.2, 22.1]†
	plurality	0.13 [0.01, 0.26]	∞; 2 [2, 3]†	∞; 9.2 [2.9, 18.0]†
OLLB, 79 top-mean models (1)	mean	1.00	5	43.4
	geometric mean	0.95	4	36.1; 35.3†
	harmonic mean	0.81	3	23.6; 20.4†
	median	0.74	2	14.5; 8.4†
	min	0.76	2	11.1; 8.4†
	Borda	1.00	3	16.7
	Condorcet winner	0.65	5; 4†	25.7; 21.9†
	plurality	0.77	2	13.3; 10.9†
HELM, 6 random subjects (100)	mean	0.35 [0.10, 0.62]	∞ [5, ∞]; 4 [3, 5]†	∞ [40.7, ∞]; 34.7 [23.0, 44.8]†
	geometric mean	0.31 [0.10, 0.62]	∞ [4, ∞]; 4 [2, 5]†	∞ [37.7, ∞]; 32.3 [19.0, 43.7]†
	harmonic mean	0.29 [0.09, 0.64]	∞ [4, ∞]; 4 [2, 5]†	∞ [37.0, ∞]; 30.8 [18.6, 43.1]†
	median	0.46 [0.13, 0.69]	∞ [3, ∞]; 3 [1, 3]†	∞ [15.2, ∞]; 13.6 [8.8, 18.0]†
	min	0.13 [0.01, 0.81]	∞ [1, ∞]; 2 [0, 4]†	∞ [4.8, ∞]; 7.8 [0, 14.9]†
	Borda	0.32 [0.13, 0.59]	∞ [6, ∞]; 5 [4, 5]†	∞ [30.8, ∞]; 20.6 [12.0, 28.0]†
	Condorcet winner	0.41 [0.12, 0.64]	∞ [4, ∞]; 4 [3, 4]†	∞ [20.8, ∞]; 17.5 [10.9, 22.3]†
	plurality	0.59 [0.22, 0.86]	2 [2, ∞]; 2 [2, 3]†	13.6 [6.3, ∞]; 9.1 [4.6, 13.5]†

Table 9: Median costs of the five score rules at the ceiling cap: tasks as a fraction of tasks (K/n), points (L_1) and largest rise (L_∞) in raw points and in units of each task's standard deviation over all models of the frame (separately optimised, ddof 0). Source: `analysis/results/final_costscale_summary.csv`.

frame (N)	rule	K/n	L_1	L_1/n	$L_{1,\mathrm{sd}}$	L_∞	$L_{\infty,\mathrm{sd}}$
HELM (78)	mean	0.28	661.0 [696.5]	11.6 [12.2]	49.64 [52.35]	12.6 [13.2]	1.10 [1.15]
	geometric mean	0.28	633.4 [668.7]	11.1 [11.7]	50.52 [53.24]	13.8 [14.2]	1.19 [1.23]
	harmonic mean	0.28	609.0 [643.6]	10.7 [11.3]	50.40 [53.02]	15.2 [15.3]	1.30 [1.32]
	median	0.37	110.8 [112.3]	1.9 [2.0]	9.89 [10.28]	9.8	0.91
	min	0.19 [0.20]	97.5 [98.1]	1.7	8.23 [8.39]	22.2	1.59
OLLB (4495)	mean	0.33	180.5	30.1	11.13	30.1	2.45
	geometric mean	0.50	153.2	25.5	15.40	27.8	2.67
	harmonic mean	0.50	124.8	20.8	14.41	24.9	2.90
	median	0.50	105.9	17.7	7.23	36.2	2.53
	min	0.50	43.2	7.2	6.41	17.9	3.39
OLLB top-500 (499)	mean	0.17	71.1	11.8	7.33	11.8	1.91
	geometric mean	0.17	48.5	8.1	9.95	11.0	2.01
	harmonic mean	0.33	31.9	5.3	7.31	9.9	1.99
	median	0.33	26.5	4.4	4.57	10.9	1.85
	min	0.33	12.3	2.1	2.93	7.6	1.89

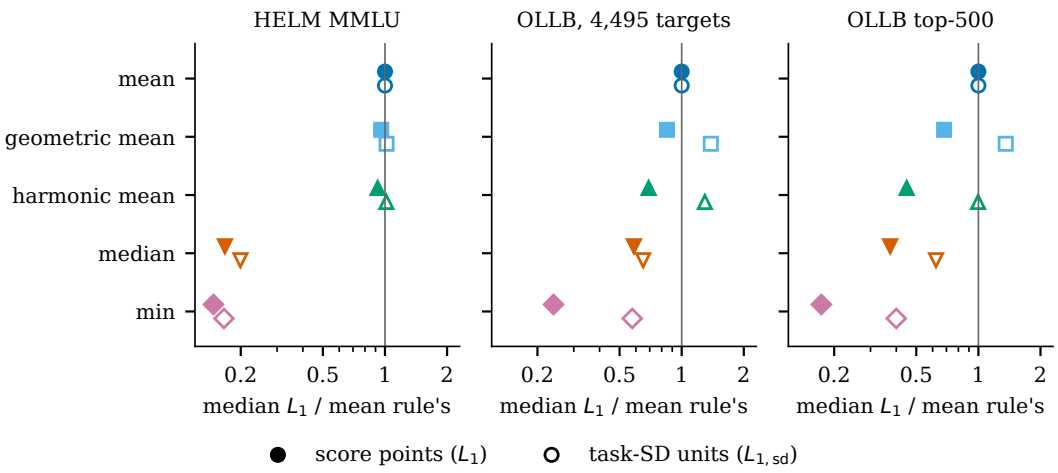


Figure 4: The mean rule is not the most expensive score rule once costs are measured in task-SD units. Median total investment L_1 to make each of the mean-non-leading targets first at the ceiling cap, relative to the mean rule's median, in raw score points (filled) and with each task's rise divided by that task's standard deviation over all models of the frame (open; separately optimised, not a rescaling of the point optimum). Log axis, 1 = mean. On full OLLB the median $L_{1,\mathrm{sd}}$ is 11.1 for the mean rule against 15.4 geometric, 14.4 harmonic, 7.2 median and 6.4 minimum; on HELM it is 49.6 against 50.5, 50.4, 9.9 and 8.2. The point-unit ordering (mean most expensive) and the SD-unit ordering therefore differ. SD units are frame dependent, not training costs. Medians over all targets of fixed censuses; table: values in both units. Source: `figures/fig15_units.provenance.json`.

B FURTHER RESULTS

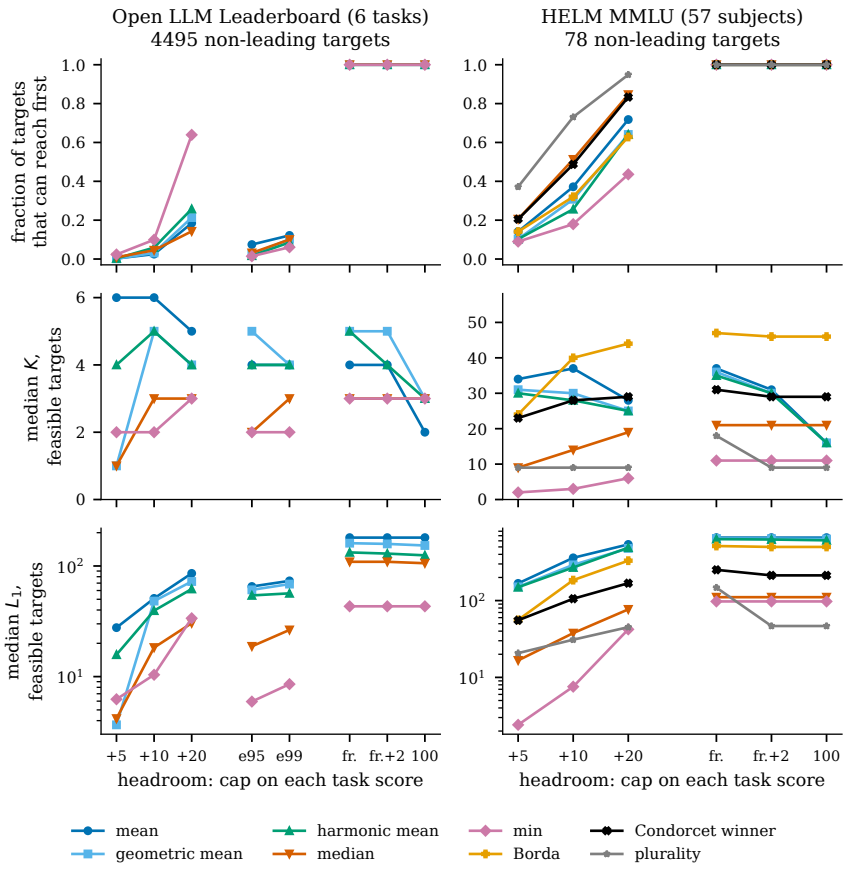


Figure 5: Cap sensitivity (all non-leading targets). (a) feasible fraction; (b, c) medians of K and L_1 over feasible targets only, so the target set differs between rules and caps. Caps: $+h$ points per task; e95/e99 empirical gain quantiles (OLLB only, descriptive); fr.: best existing score on each task (+2); 100: ceiling. Missing points: rule not run under that cap. Source: `figures/fig2_caps.provenance.json`.

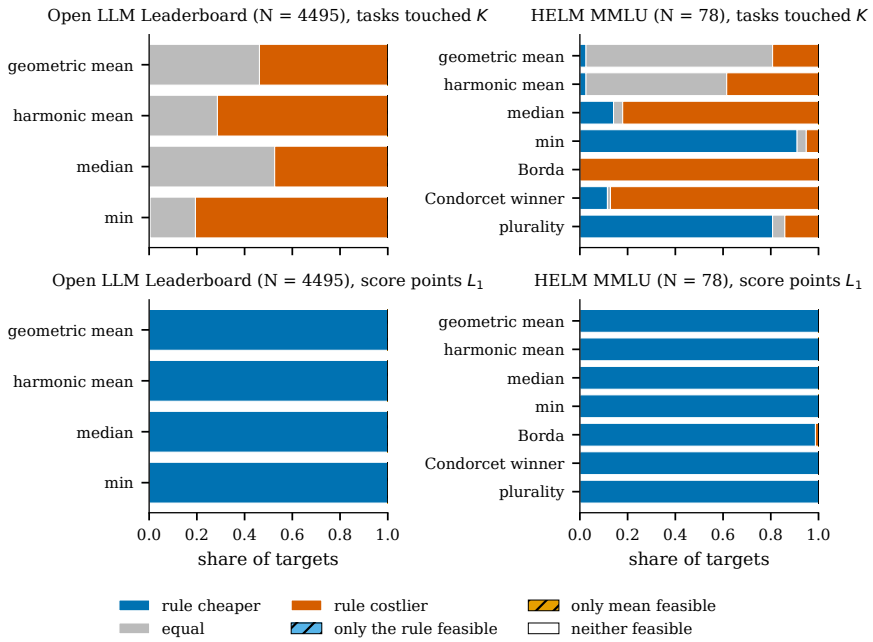


Figure 6: Same-target comparison with the mean rule under the ceiling cap (non-leading targets): share of targets for which the rule is cheaper, equal or costlier than the mean among targets feasible under both. The mean rule is the reference and has no row. Differences mix each rule's gap with its response to increases (the original leader differs), so this is not a gap-matched comparison. Source: `figures/fig3_paired_ceil.provenance.json`.

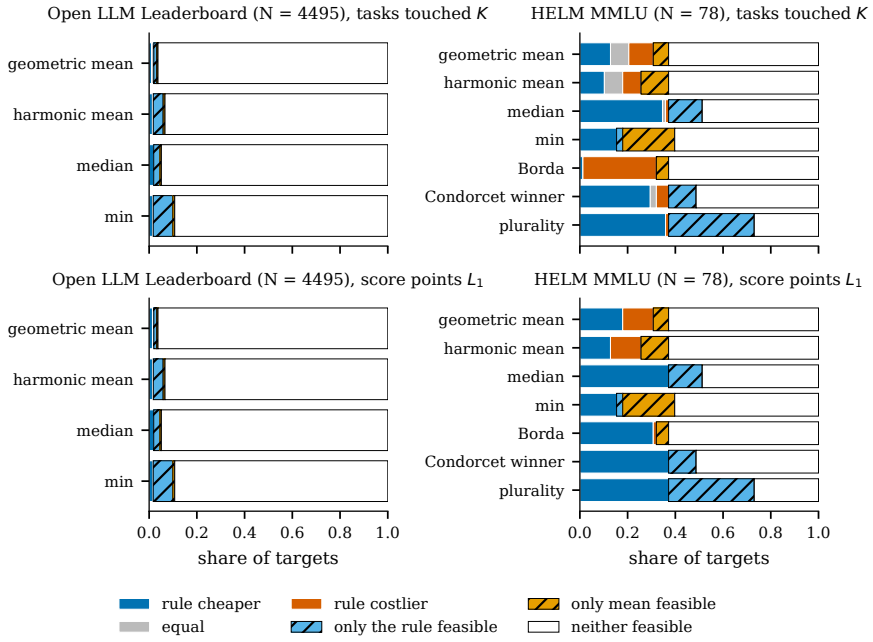


Figure 7: As Figure 6 under the 10-point cap; hatched: feasible under only one of the two rules, white: under neither. Source: `figures/fig3_paired_gain10.provenance.json`.

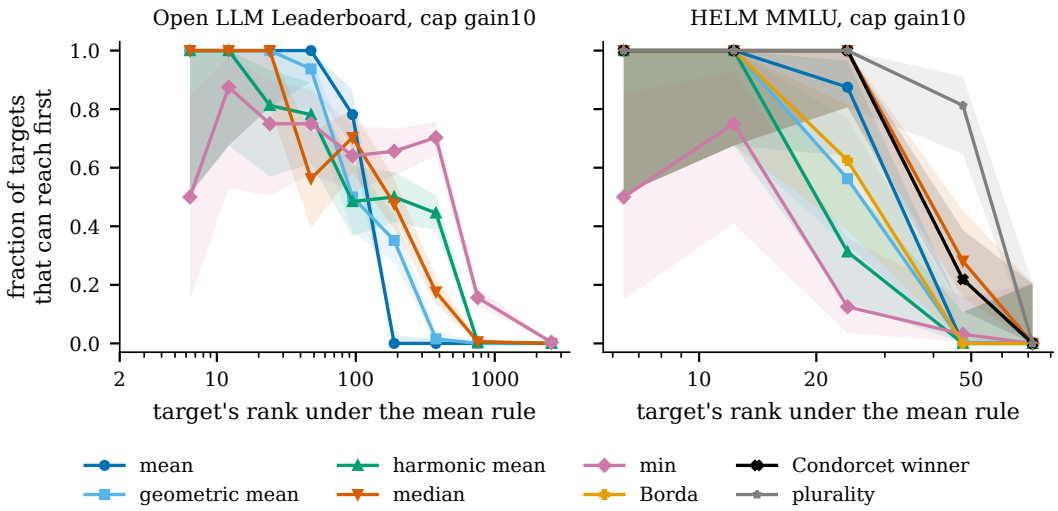


Figure 8: Fraction of targets that can become first at all under the 10-point cap, by rank under the mean; Wilson 95% bands (fixed census). Source: `figures/fig1b_feasible_by_rank.provenance.json`.

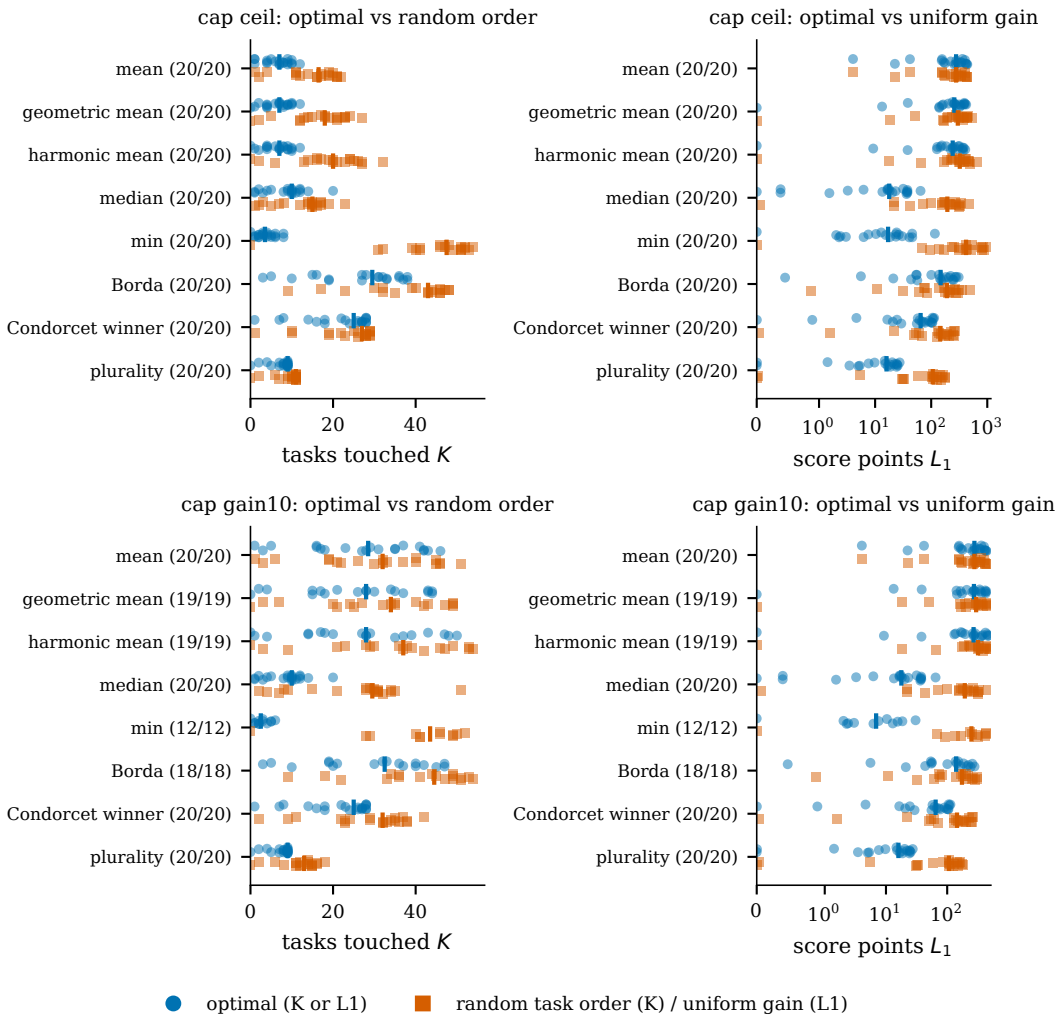


Figure 9: Targeting controls on HELM (20 targets of mean rank 2–21): optimal K versus random task order (200 orders), and optimal L_1 versus the uniform clipped improvement at the optimal L_∞. Source: `figures/fig6_controls.provenance.json`.

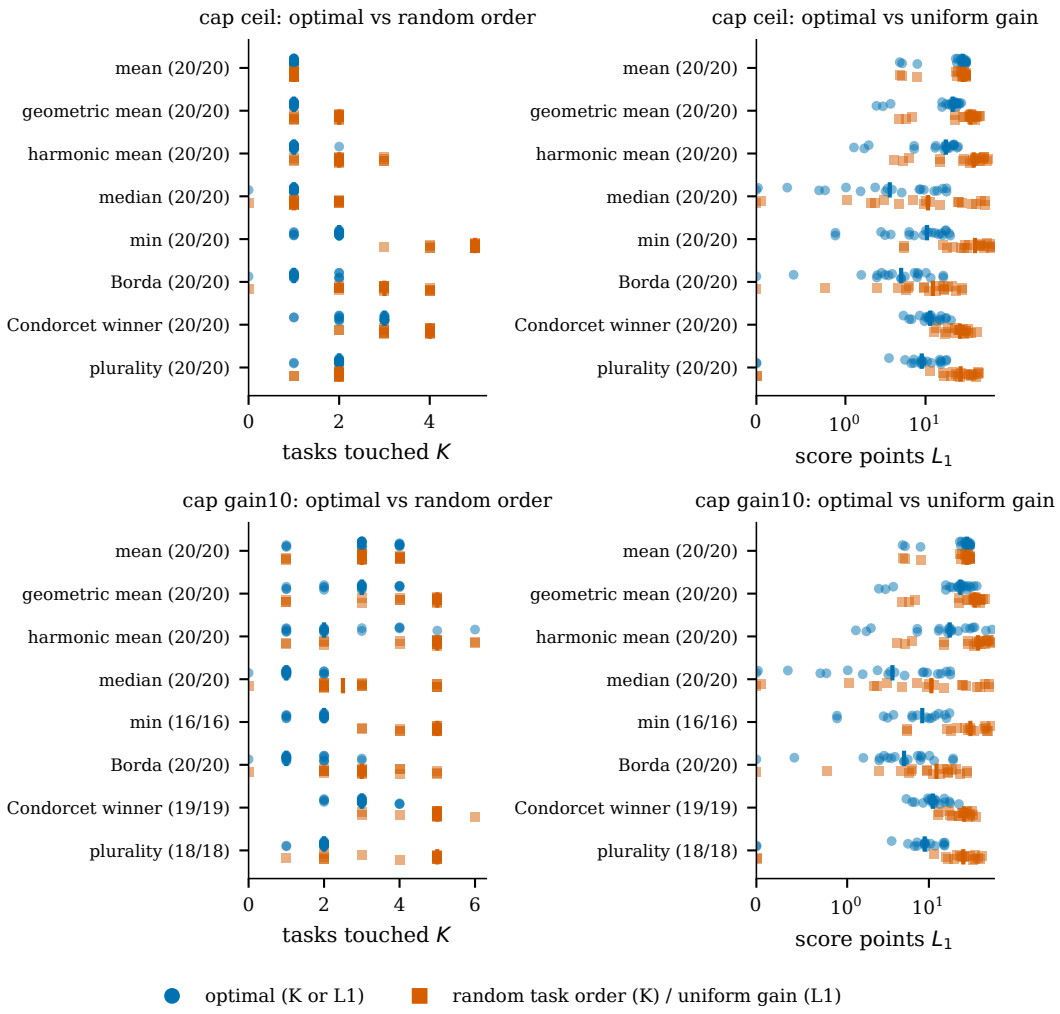


Figure 10: As Figure 9 on the Open LLM Leaderboard top-500 frame (20 targets of mean rank 2–21 within the frame; the frames are not pooled). Source: `figures/fig6c_top500_controls.provenance.json`.

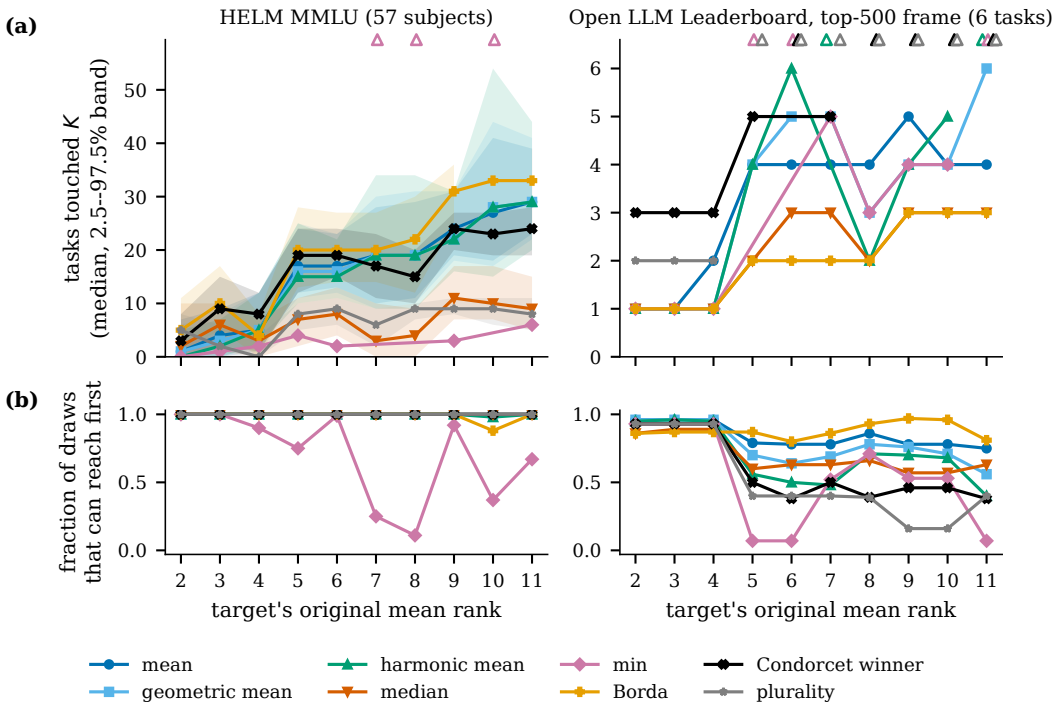


Figure 11: Task-mixture sensitivity (100 resamplings of the tasks with replacement, cap +10, seed 0) on HELM MMLU (57 subjects) and on the Open LLM Leaderboard top-500 frame (6 tasks); ten fixed targets of original mean rank 2–11 within each frame, not pooled. Bands are 2.5–97.5 percentiles, a sensitivity range, not a confidence interval. Source: `figures/fig4_robustness.provenance.json`.

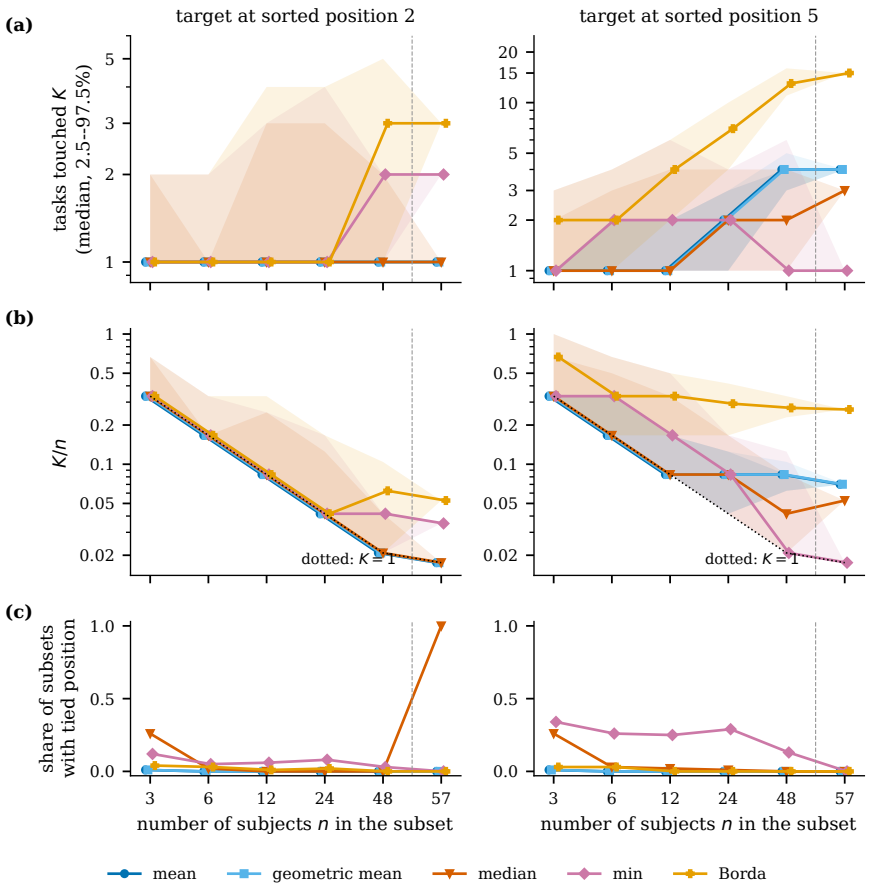


Figure 12: Task-count sensitivity on HELM MMLU: 100 random subsets of n subjects per size, ceiling cap; the target is the model at sorted position 2 or 5 under each rule within each subset, so it differs between rules and subsets. At $n = 57$ the subsets differ only by permutation. A sensitivity range, not a confidence interval and not evidence that datasets differ because of their task counts. Source: `figures/fig7_taskcount.provenance.json`.

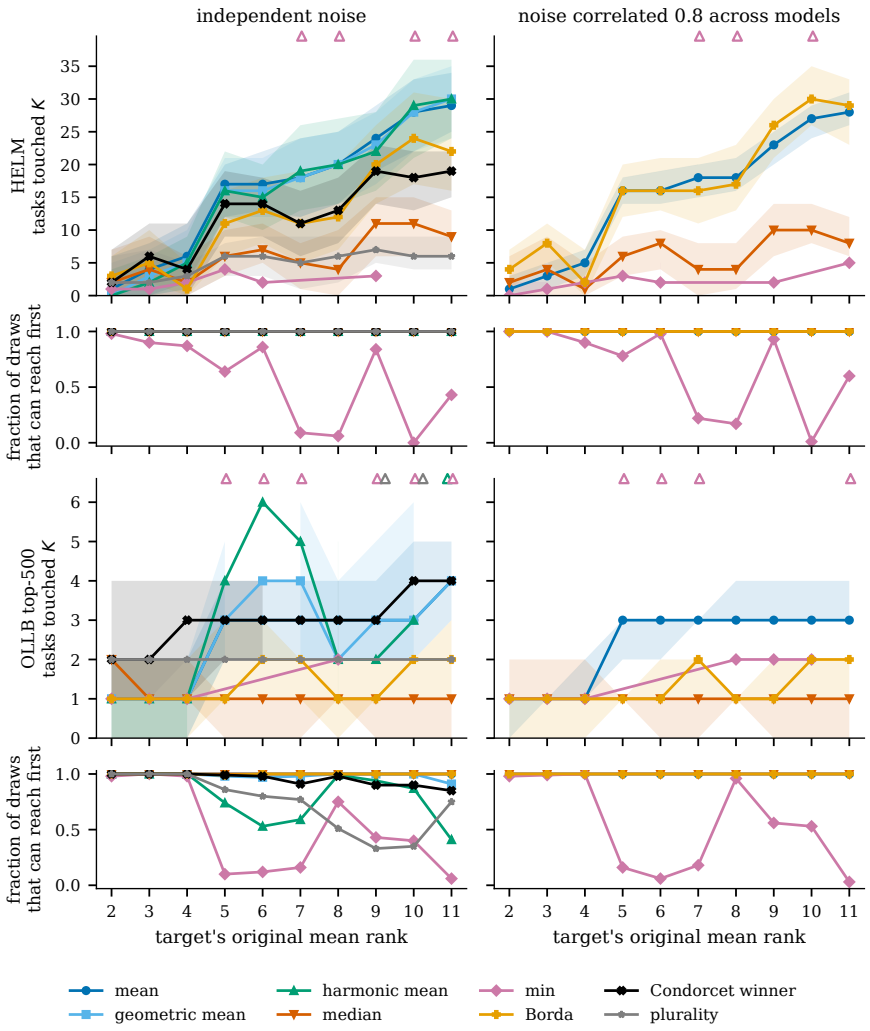


Figure 13: Sensitivity to assumed Gaussian score noise (independent across models, or correlated 0.8 within a task), cap +10, 100 draws; the ten fixed targets of original mean rank 2–11 in each frame. Plug-in standard errors, so a sensitivity range under an assumption, not a measurement-error interval. Source: `figures/fig8_noise.provenance.json`.

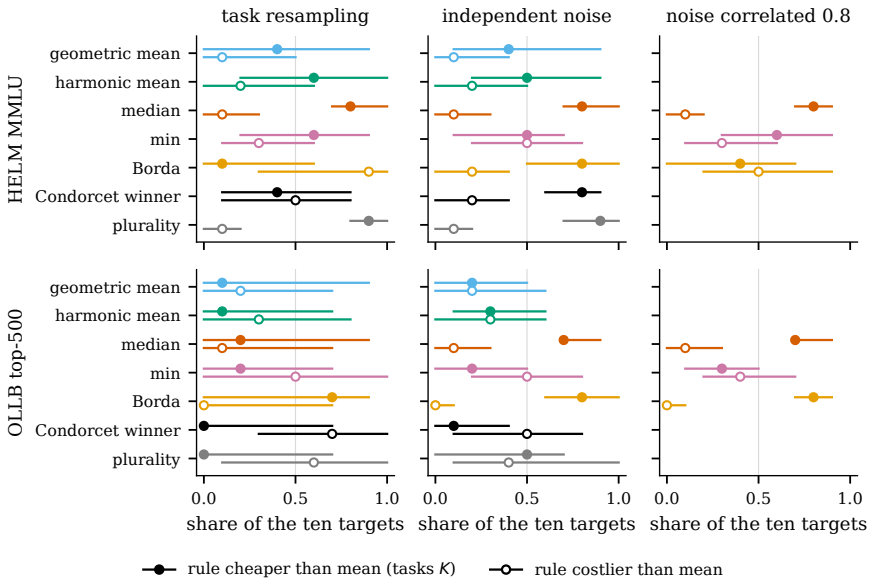


Figure 14: Same-draw comparison with the mean rule under task resampling and score noise, by tasks touched K (ten fixed targets, cap $+10$): share of targets for which the rule needs fewer (filled) or more (open) tasks than the mean on that draw. Ranges over draws, not confidence intervals. The corresponding figure for L_1 is Figure 15. Source: `figures/fig9_paired_perturb_K.provenance.json`.

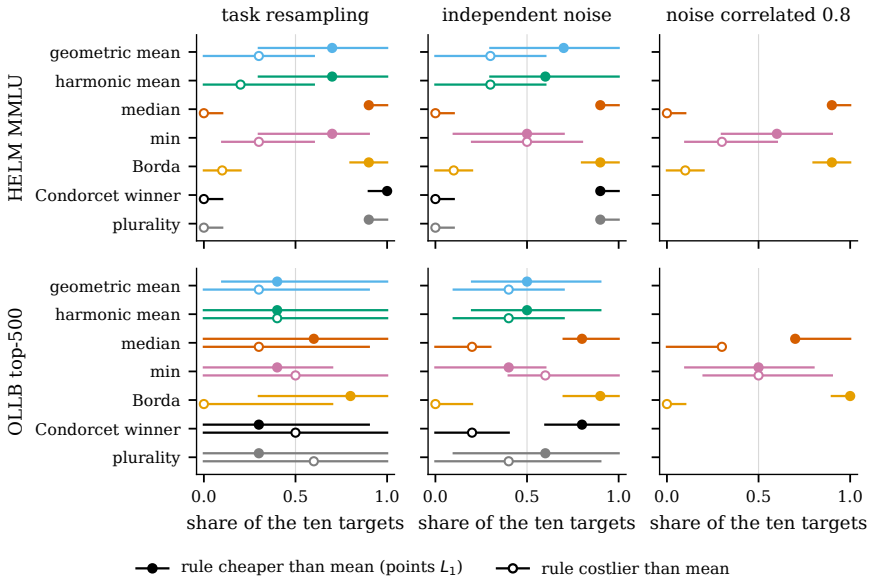


Figure 15: As Figure 14 for total points L_1. Source: `figures/fig9_paired_perturb_L1.provenance.json`.

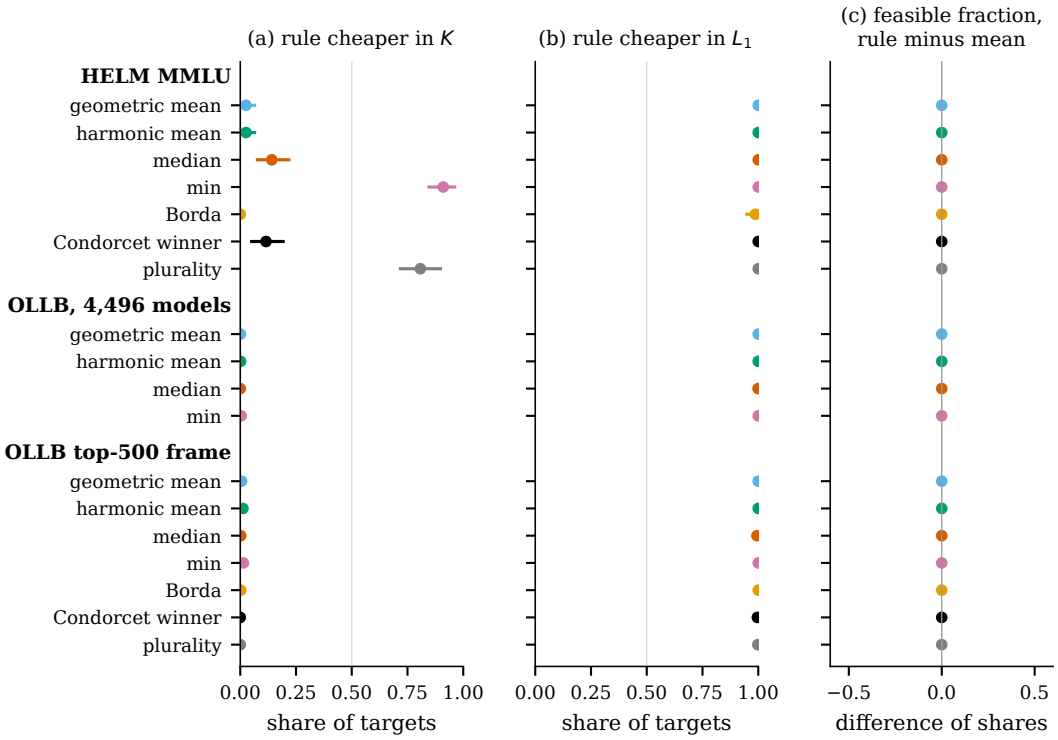


Figure 16: Target-resampling ranges for the same-target comparison with the mean rule under the ceiling cap (1000 resamplings of the targets, seed 23): (a, b) share of targets for which the rule needs fewer tasks (points) than the mean; (c) difference in feasible fraction. Sensitivity ranges of fixed censuses, not population confidence intervals; frames are not pooled. Source: `figures/fig10_paired_bootstrap_ceil.provenance.json`.

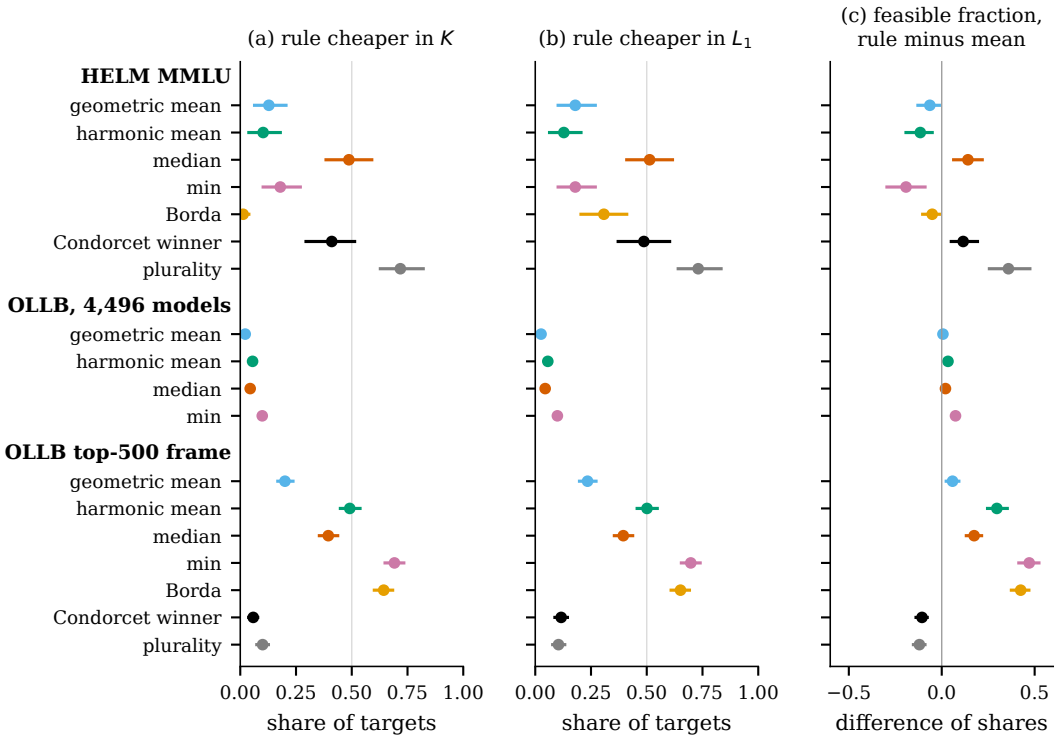


Figure 17: As Figure 16 under the 10-point cap. Source: `figures/fig10_paired_bootstrap_gain10.provenance.json`.

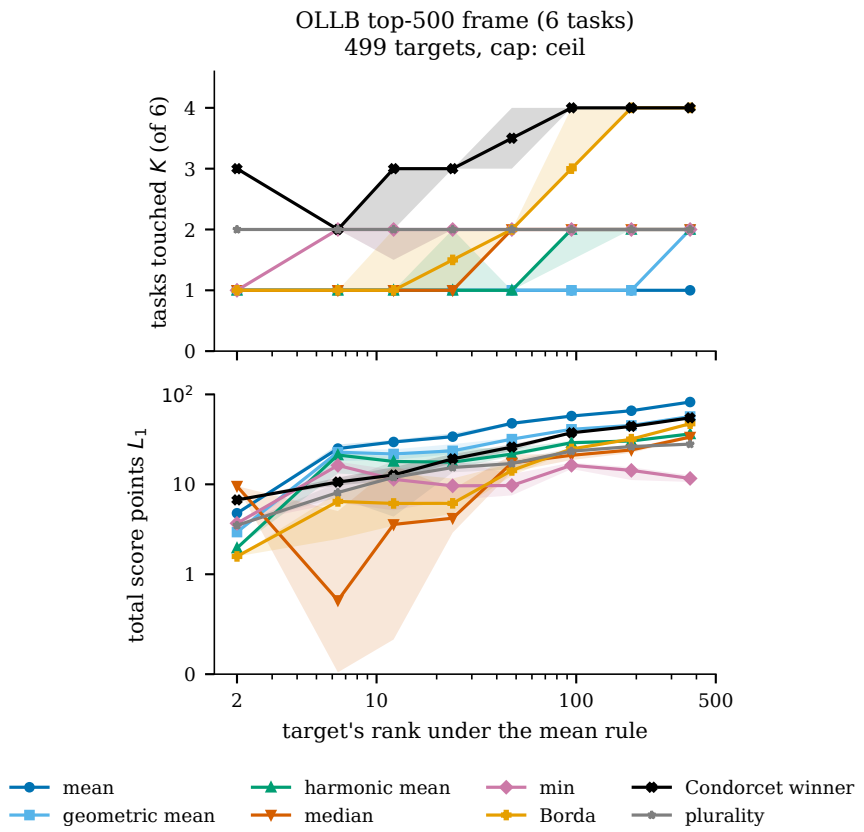


Figure 18: Minimum investment versus the target's rank under the mean within the top-500 frame (ceiling cap), per rule including the ordinal rules; the frame is not comparable with the 4,496-model census. Source: `figures/fig1c_top500_rank_rules_ceil.provenance.json`.

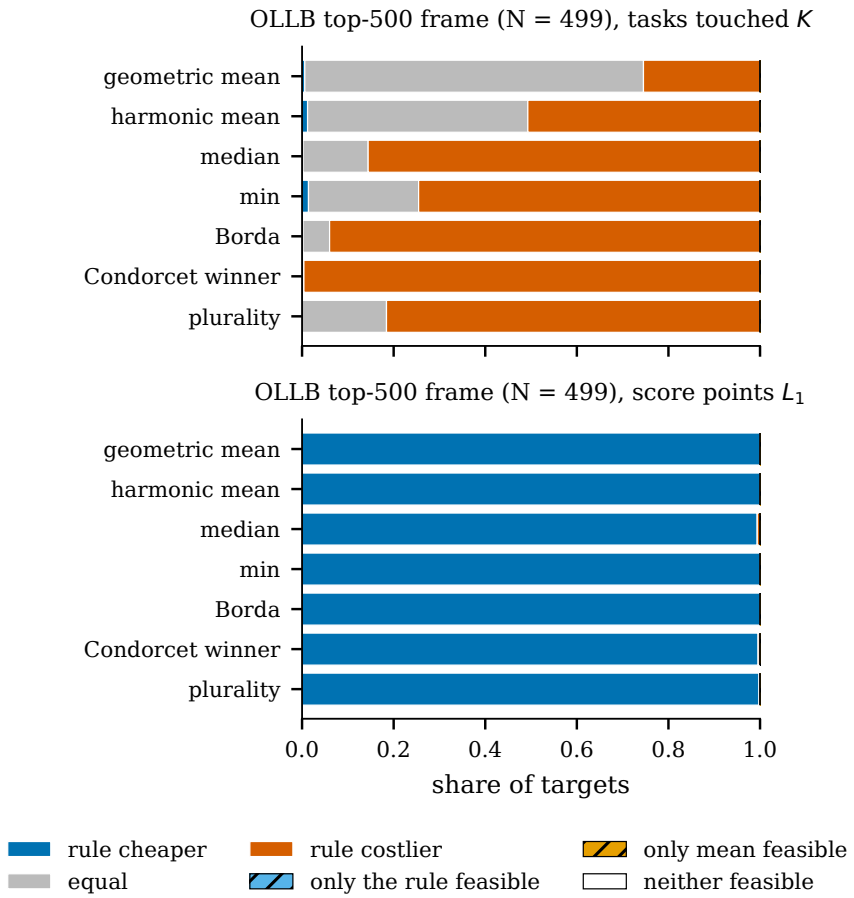


Figure 19: Same-target comparison with the mean on the top-500 frame, ceiling cap (conventions of Figure 6). The 10-point cap version is Figure 20. Source: `figures/fig3c_top500_paired_ceil.provenance.json`.

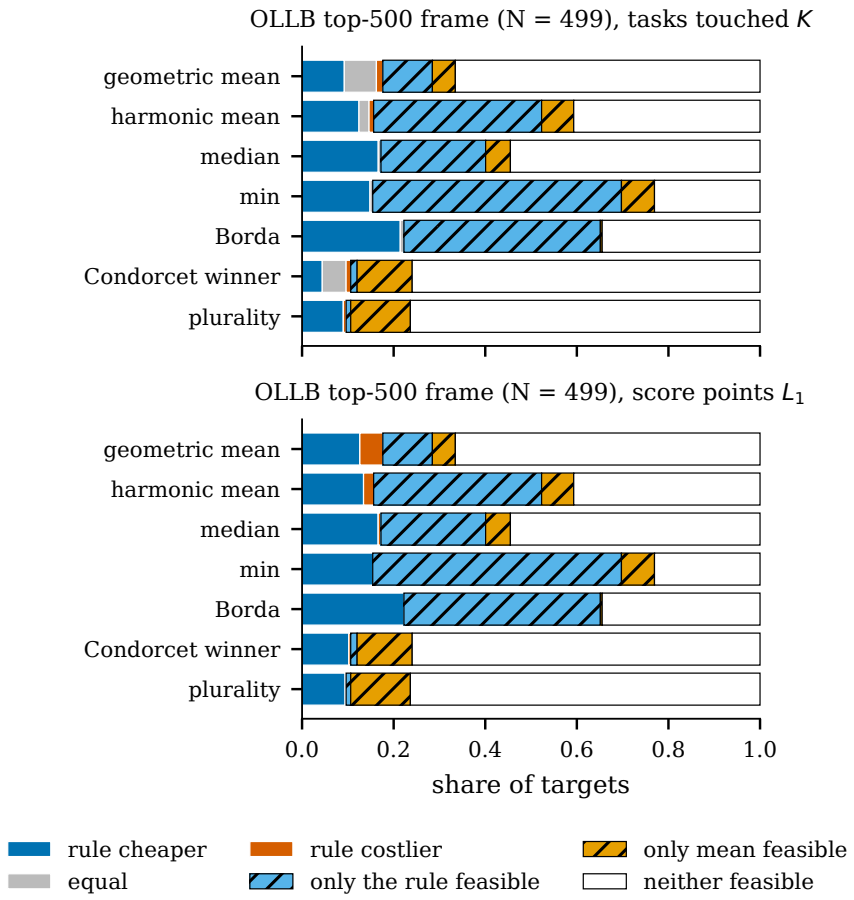


Figure 20: As Figure 19 under the 10-point cap. Source: `figures/fig3c_top500_paired_gain10.provenance.json`.

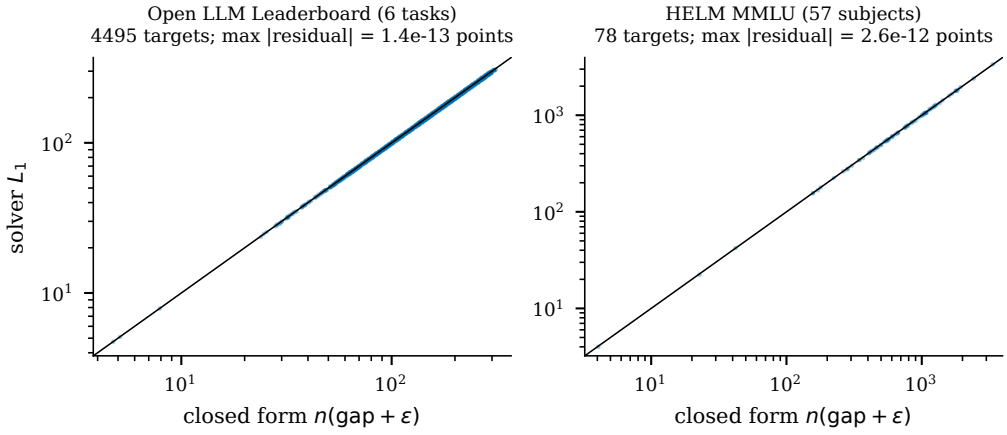


Figure 21: Solver minimum L_1 for the mean against the closed form $n(\tau + \varepsilon - \bar{x})$ (Section 3, (i)); the line is the identity. Source: `figures/fig5_theory_check.provenance.json`.

SENSITIVITY: FURTHER DETAIL

Appendix Fig. 11 follows ten fixed targets (mean rank 2–11 within each frame) under 100 resamplings of the tasks (cap $+10$). On HELM the number of tasks varies with the draw; the median

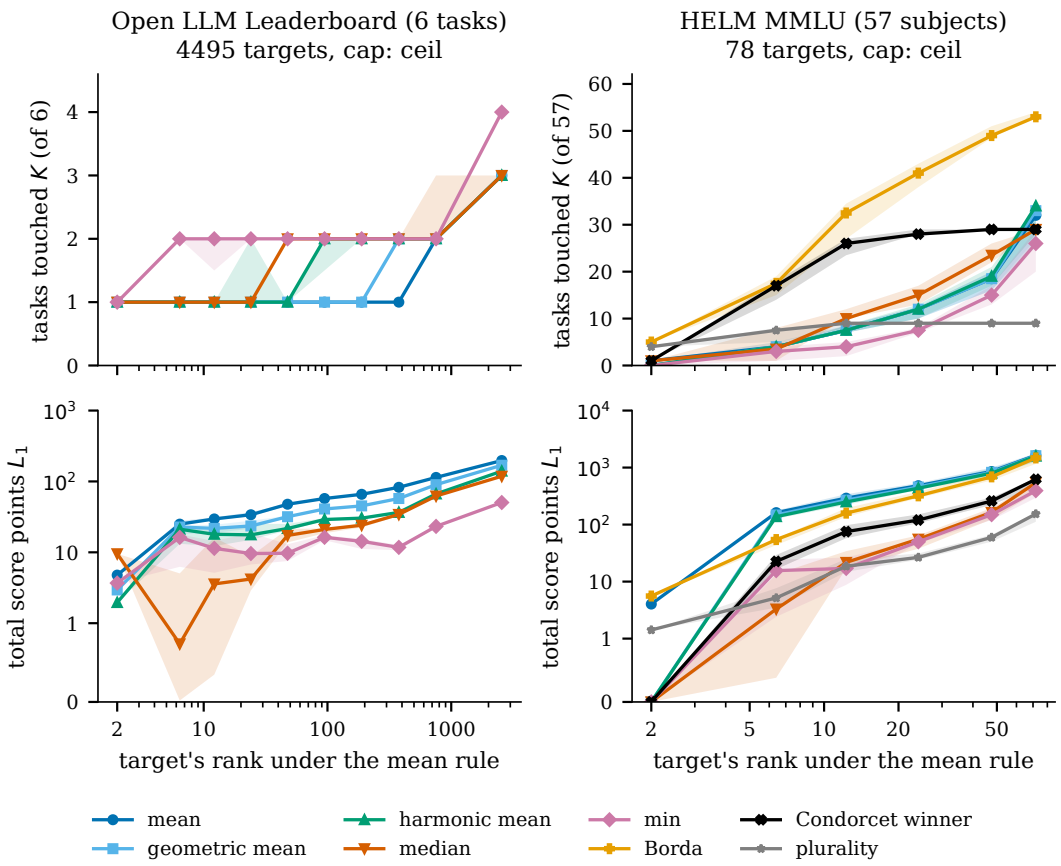


Figure 22: Investment versus the target's rank under the mean (ceiling cap): bin medians with 95% bootstrap bands over targets in each bin (fixed census, not a population). Ordinal rules only on HELM. Source: `figures/fig1_rank_rules_ceil.provenance.json`.

draw needs fewer tasks under the median rule than under the mean for 8 of the 10 targets; under the minimum rule, whether first place is reachable at all varies strongly across draws (fraction of draws feasible between 0.11 and 1.0 depending on the target, against at least 0.88 for every other rule). On the top-500 Open LLM Leaderboard frame the share of the ten followed targets that both the minimum rule and the mean can win within a draw is lower than on HELM (median over task resamplings 0.4, against 0.7 on HELM). Score-noise perturbations (independent and correlated) and the task-count experiment on HELM ($n = 3, \ldots, 57$ subjects, 100 subsets each) were also run; their percentile ranges are sensitivity ranges under the assumptions of Section 4, not confidence intervals, and are shown in Appendix Figs. 12, 13 and 14 (values in analysis/results/summary/). In the task-count experiment the targets are the models at sorted positions 2 and 5 of each random subset, reselected separately for every rule and subset (possible ties are counted in the supplement), so it does not follow one fixed model as n grows and implies no scaling law; for example, the median K of the mean rule for position 2 is 1 at every n from 3 to 57 subjects tested, while its median L_1 rises from 1.7 to 7.3 points between $n = 3$ and $n = 48$ and is 4.0 at $n = 57$, where every subset contains all 57 subjects.

Margin and competitor frame. Two further checks on the 20 followed targets per frame (original mean rank 2–21; Appendix Tables 21 and 22; analysis/docs/REVIEW_FINDINGS.md) show that the costs are conditional on conventions. Changing the winning margin ε from 0.001 to 0.1 changed no feasibility for the mean or Borda at the ceiling or 10-point cap, but changed the task count K of the mean for 2/20 targets at the ceiling and 7/20 at the 10-point cap on HELM (2/20 on the Open LLM Leaderboard), and of Borda for 1/20 on HELM at the 10-point cap; the mean's L_1

rises by $n \times 0.099$ as the closed form predicts (5.64 points on HELM, 0.59 on the Open LLM Leaderboard). Removing the mean's leader from the rivals (all rules, 10-point cap, original mean rank 2–11) lowered the optimal Borda K for 10/10 HELM targets and the mean's for 5/10, but the minimum rule's for 0/10; under ten random-half rival frames, the share of the ten targets with lower Borda K has median 1.0 (range 0.1–1.0), against 0.5 (0–1.0) for the mean. On this HELM near-top panel, Borda's task count therefore responds more strongly to the rival set than the mean's does. These are near-top panels with ten frames, descriptive ranges and not summaries over all targets.

HEADLINE RANGES, MEDIAN CONVENTIONS, CONTROLS IN POINTS AND PROCESSING CHOICES

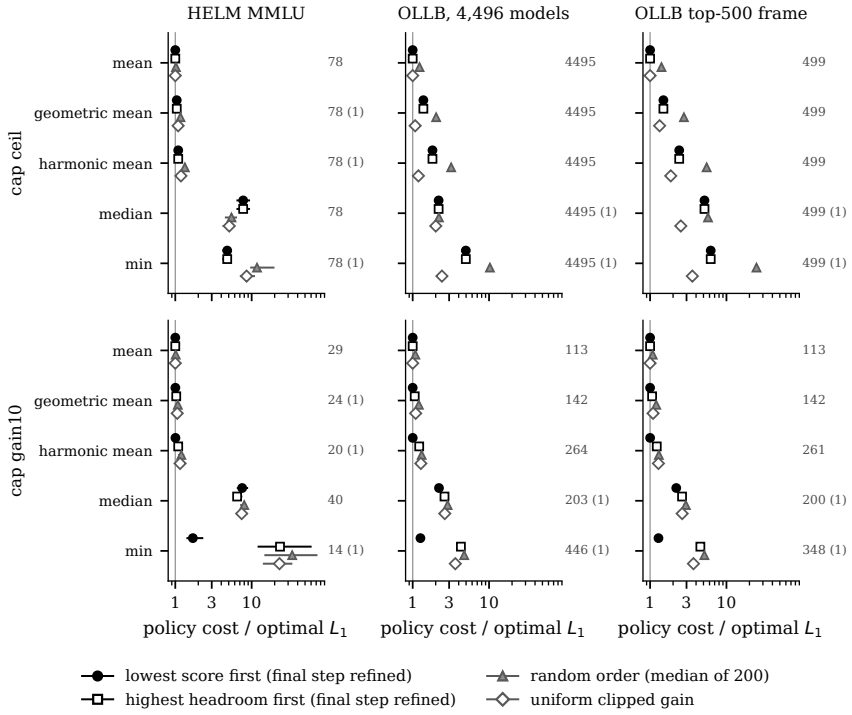


Figure 23: As Figure 1 for total points L_1 (the greedy policies lift each touched task to its cap, the last step refined, so these ratios describe those named policies). Source: `figures/fig11_policy_controls_L1.provenance.json`.

Table 10: Table 2 with 2.5–97.5 percentile ranges from 1000 resamplings of the targets (seed 23), ceiling cap: median over all non-leading targets (inverted-CDF convention). Descriptive ranges for the fixed censuses, not population confidence intervals. Source: `analysis/results/final_headline_intervals.csv`.

rule	feas. [range]	K [range]	L_1 [range]
HELM MMLU (57 subjects), N = 78 non-leading targets			
mean	78/78 [78, 78]	16 [13, 18]	661 [560, 842]
geometric mean	78/78 [78, 78]	16 [13, 18]	633 [533, 814]
harmonic mean	78/78 [78, 78]	16 [13, 18]	609 [509, 788]
median	78/78 [78, 78]	21 [18, 23]	111 [73.8, 158]
min	78/78 [78, 78]	11 [9, 15]	97.5 [55.8, 145]
Borda	78/78 [78, 78]	46 [44, 49]	500 [397, 678]
Condorcet winner	78/78 [78, 78]	29 [29, 29]	213 [159, 247]
plurality	78/78 [78, 78]	9 [9, 9]	46.7 [33.8, 58.1]
OLLB (6 tasks, 4,496 models), N = 4495 non-leading targets			
mean	4495/4495 [4495, 4495]	2 [2, 2]	181 [178, 182]
geometric mean	4495/4495 [4495, 4495]	3 [3, 3]	153 [151, 155]
harmonic mean	4495/4495 [4495, 4495]	3 [3, 3]	125 [124, 126]
median	4495/4495 [4495, 4495]	3 [3, 3]	106 [105, 107]
min	4495/4495 [4495, 4495]	3 [3, 3]	43.2 [42.4, 44.1]
OLLB top-500 frame (6 tasks), N = 499 non-leading targets			
mean	499/499 [499, 499]	1 [1, 1]	71.1 [69.3, 73.4]
geometric mean	499/499 [499, 499]	1 [1, 1]	48.5 [47.8, 49.7]
harmonic mean	499/499 [499, 499]	2 [2, 2]	31.9 [31.2, 32.7]
median	499/499 [499, 499]	2 [2, 2]	26.5 [25.5, 27.5]
min	499/499 [499, 499]	2 [2, 2]	12.3 [11.4, 13.2]
Borda	499/499 [499, 499]	4 [4, 4]	36.4 [34.5, 38.1]
Condorcet winner	499/499 [499, 499]	4 [4, 4]	47.0 [45.8, 48.4]
plurality	499/499 [499, 499]	2 [2, 2]	26.3 [25.8, 26.6]

Table 11: As Table 10 under the 10-point cap; ∞: median over all targets (infeasible targets included); †: median over feasible targets only.

rule	feas. [range]	K [range]	L_1 [range]
HELM MMLU (57 subjects), N = 78 non-leading targets			
mean	29/78 [20, 38]	∞; $37^\dagger$ [28, 46]	∞; $360^\dagger$ [276, 440]
geometric mean	24/78 [16, 32]	∞; $30^\dagger$ [18, 43]	∞; $292^\dagger$ [209, 417]
harmonic mean	20/78 [13, 28]	∞; $28^\dagger$ [17, 37]	∞; $270^\dagger$ [168, 364]
median	40/78 [32, 49]	21 [18, ∞]; $14^\dagger$	114 [68.2, ∞]; $37.7^\dagger$
min	14/78 [8, 21]	∞; $3^\dagger$ [2, 5]	∞; $7.6^\dagger$ [2.4, 15.5]
Borda	25/78 [17, 33]	∞; $40^\dagger$ [30, 47]	∞; $185^\dagger$ [100, 280]
Condorcet winner	38/78 [29, 47]	∞ [29, ∞]; $28^\dagger$	∞ [159, ∞]; $106^\dagger$
plurality	57/78 [49, 65]	9 [9, 10]; $9^\dagger$	46.7 [32.5, 61.0]; $31.0^\dagger$
OLLB (6 tasks, 4,496 models), N = 4495 non-leading targets			
mean	113/4495 [92, 135]	∞; $6^\dagger$ [5, 6]	∞; $50.9^\dagger$ [47.5, 52.8]
geometric mean	142/4495 [120, 167]	∞; $5^\dagger$ [5, 5]	∞; $48.2^\dagger$ [43.9, 49.9]
harmonic mean	264/4495 [234, 298]	∞; $5^\dagger$ [4, 5]	∞; $39.5^\dagger$ [37.6, 41.1]
median	203/4495 [176, 232]	∞; $3^\dagger$ [3, 3]	∞; $18.3^\dagger$ [16.5, 20.4]
min	446/4495 [406, 487]	∞; $2^\dagger$ [2, 2]	∞; $10.4^\dagger$ [9.8, 11.0]
OLLB top-500 frame (6 tasks), N = 499 non-leading targets			
mean	113/499 [95, 132]	∞; $6^\dagger$ [5, 6]	∞; $50.9^\dagger$ [47.5, 52.6]
geometric mean	142/499 [122, 163]	∞; $5^\dagger$ [5, 5]	∞; $48.2^\dagger$ [43.9, 49.9]
harmonic mean	261/499 [240, 282]	6 [6, ∞]; $5^\dagger$	58.1 [54.2, ∞]; $39.4^\dagger$
median	200/499 [178, 220]	∞; $3^\dagger$ [3, 3]	∞; $18.3^\dagger$ [16.4, 20.1]
min	348/499 [328, 368]	2 [2, 2]; $2^\dagger$	12.3 [11.4, 13.3]; $9.6^\dagger$
Borda	325/499 [306, 345]	6 [5, 6]; $4^\dagger$	39.9 [36.4, 41.9]; $30.6^\dagger$
Condorcet winner	60/499 [47, 74]	∞; $4^\dagger$ [4, 5]	∞; $23.3^\dagger$ [20.3, 25.7]
plurality	53/499 [40, 67]	∞; $2^\dagger$ [2, 2]	∞; $13.1^\dagger$ [10.9, 14.6]

Table 12: Feasible non-leading targets per score rule, frame and cap, with the number of targets whose optimal cost is zero in parentheses (these enter the ratios in Fig. 1 as defined in the source file). Source: `analysis/results/final_controls_summary.csv`.

	HELM		OLLB		top-500	
rule	ceil	gain10	ceil	gain10	ceil	gain10
mean	78	29	4495	113	499	113
geometric mean	78 (1)	24 (1)	4495	142	499	142
harmonic mean	78 (1)	20 (1)	4495	264	499	261
median	78	40	4495 (1)	203 (1)	499 (1)	200 (1)
min	78 (1)	14 (1)	4495 (1)	446 (1)	499 (1)	348 (1)

Table 13: Entries where the inverted-CDF (lower-middle) and the usual midpoint median differ; the optimisation problem of every target is unchanged. Source: `analysis/results/final_headline_intervals.csv`.

frame	rule	cap	measure	inverted-CDF	[range]	midpoint	[range]
HELM	min	ceil	K	11	[9, 15]	11.5	[9, 15]
HELM	Borda	ceil	K	46	[44, 49]	47	[44, 49]
HELM	mean	ceil	L_1	661	[560, 842]	696	[560, 852]
HELM	geometric mean	ceil	L_1	633	[533, 814]	669	[533, 824]
HELM	harmonic mean	ceil	L_1	609	[509, 788]	644	[509, 798]
HELM	median	ceil	L_1	111	[73.8, 158]	112	[73.8, 158]
HELM	min	ceil	L_1	97.5	[55.8, 145]	98.1	[57.8, 149]
HELM	Borda	ceil	L_1	500	[397, 678]	534	[397, 687]
HELM	Condorcet winner	ceil	L_1	213	[159, 247]	213	[162, 253]
HELM	plurality	ceil	L_1	46.7	[33.8, 58.1]	47.2	[34.5, 59.6]
HELM	geometric mean	gain10	K	$30^{\dagger}$	[18, 43]	$32^{\dagger}$	[21, 43]
HELM	geometric mean	gain10	L_1	$292^{\dagger}$	[209, 417]	$315^{\dagger}$	[209, 421]
HELM	harmonic mean	gain10	L_1	$270^{\dagger}$	[168, 364]	$273^{\dagger}$	[173, 371]
HELM	median	gain10	L_1	114	[68.2, ∞]	116	[71.0, ∞]
HELM	min	gain10	L_1	$7.6^{\dagger}$	[2.4, 15.5]	$9.0^{\dagger}$	[2.6, 18.5]
HELM	Condorcet winner	gain10	L_1	$106^{\dagger}$	[77.2, 134]	$107^{\dagger}$	[83.8, 141]
HELM	plurality	gain10	L_1	46.7	[32.5, 61.0]	47.2	[33.8, 61.5]
OLLB	geometric mean	gain10	L_1	$48.2^{\dagger}$	[43.9, 49.9]	$48.4^{\dagger}$	[44.3, 50.1]
OLLB	harmonic mean	gain10	L_1	$39.5^{\dagger}$	[37.6, 41.1]	$39.5^{\dagger}$	[37.8, 41.2]
OLLB top-500	geometric mean	gain10	L_1	$48.2^{\dagger}$	[43.9, 49.9]	$48.4^{\dagger}$	[44.2, 49.9]
OLLB top-500	min	gain10	L_1	$9.6^{\dagger}$	[9.2, 10.1]	$9.6^{\dagger}$	[9.2, 10.1]
OLLB top-500	Condorcet winner	gain10	L_1	$23.3^{\dagger}$	[20.3, 25.7]	$23.4^{\dagger}$	[20.7, 25.7]

Table 14: Share of targets for which each other rule is cheaper than the mean (direction: other rule cheaper), with target-resampling ranges. Source: `analysis/results/final_headline_shares.csv`.

	cap ceil		cap gain10	
rule	K	L_1	K	L_1
HELM MMLU (57 subjects), N = 78 non-leading targets				
geometric mean	0.03 [0.00, 0.06]	1.00 [1.00, 1.00]	0.13 [0.05, 0.21]	0.18 [0.10, 0.27]
harmonic mean	0.03 [0.00, 0.06]	1.00 [1.00, 1.00]	0.10 [0.04, 0.17]	0.13 [0.06, 0.21]
median	0.14 [0.06, 0.22]	1.00 [1.00, 1.00]	0.49 [0.38, 0.60]	0.51 [0.41, 0.63]
min	0.91 [0.85, 0.97]	1.00 [1.00, 1.00]	0.18 [0.10, 0.27]	0.18 [0.10, 0.27]
Borda	0.00 [0.00, 0.00]	0.99 [0.96, 1.00]	0.01 [0.00, 0.04]	0.31 [0.21, 0.41]
Condorcet winner	0.12 [0.05, 0.19]	1.00 [1.00, 1.00]	0.41 [0.29, 0.53]	0.49 [0.37, 0.60]
plurality	0.81 [0.72, 0.90]	1.00 [1.00, 1.00]	0.72 [0.62, 0.82]	0.73 [0.63, 0.83]
every comparator	–	0.99 [0.96, 1.00]	–	0.10 [0.04, 0.17]
OLLB (6 tasks, 4,496 models), N = 4495 non-leading targets				
geometric mean	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]	0.02 [0.02, 0.03]	0.03 [0.02, 0.03]
harmonic mean	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]	0.06 [0.05, 0.06]	0.06 [0.05, 0.06]
median	0.00 [0.00, 0.00]	1.00 [1.00, 1.00]	0.04 [0.04, 0.05]	0.04 [0.04, 0.05]
min	0.00 [0.00, 0.01]	1.00 [1.00, 1.00]	0.10 [0.09, 0.11]	0.10 [0.09, 0.11]
every comparator	–	1.00 [1.00, 1.00]	–	0.01 [0.01, 0.02]
OLLB top-500 frame (6 tasks), N = 499 non-leading targets				
geometric mean	0.01 [0.00, 0.01]	1.00 [1.00, 1.00]	0.20 [0.16, 0.24]	0.23 [0.20, 0.27]
harmonic mean	0.01 [0.00, 0.02]	1.00 [1.00, 1.00]	0.49 [0.45, 0.54]	0.50 [0.46, 0.55]
median	0.00 [0.00, 0.01]	0.99 [0.99, 1.00]	0.39 [0.35, 0.44]	0.39 [0.35, 0.44]
min	0.01 [0.01, 0.03]	1.00 [1.00, 1.00]	0.69 [0.65, 0.73]	0.70 [0.66, 0.74]
Borda	0.00 [0.00, 0.01]	1.00 [1.00, 1.00]	0.64 [0.60, 0.68]	0.65 [0.61, 0.69]
Condorcet winner	0.00 [0.00, 0.00]	1.00 [0.99, 1.00]	0.06 [0.04, 0.08]	0.12 [0.09, 0.14]
plurality	0.00 [0.00, 0.00]	1.00 [0.99, 1.00]	0.10 [0.08, 0.13]	0.10 [0.08, 0.13]
every comparator	–	0.99 [0.99, 1.00]	–	0.02 [0.01, 0.04]

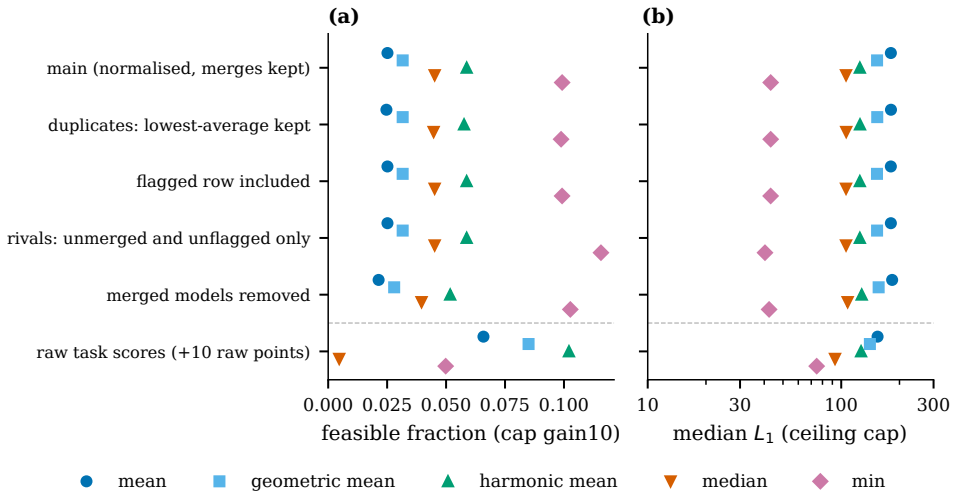


Figure 24: Open LLM Leaderboard processing choices: share of targets that can reach first place at the 10-point cap, by rule. The raw-score cap is a different intervention scale from +10 normalised points. Source: `figures/fig12_processing.provenance.json`.

Table 15: Reachable targets and ceiling-cap L_1 order under six processing choices of the Open LLM Leaderboard frame. Source: `analysis/results/final_population_summary.csv`.

choice	mean	geometric mean	harmonic mean	median	min
targets that can reach first, cap gain10 (of N non-leading)					
main (normalised, merges kept) (N = 4495)	113	142	264	203	446
duplicates: lowest-average kept (N = 4495)	111	142	259	201	444
flagged row included (N = 4496)	113	142	264	203	446
rivals: unmerged and unflagged only (N = 4495)	113	142	264	203	520
merged models removed (N = 3788)	81	106	196	150	389
raw task scores (+10 raw points) (N = 4495)	296	382	459	21	224
median L_1 over all targets, ceiling cap					
main (normalised, merges kept)	180.5	153.2	124.8	105.9	43.2
duplicates: lowest-average kept	180.7	153.5	125.0	106.1	43.2
flagged row included	180.5	153.0	124.8	105.9	43.2
rivals: unmerged and unflagged only	180.5	153.2	124.8	105.9	40.4
merged models removed	183.3	156.1	127.6	107.9	42.4
raw task scores (+10 raw points)	154.1	140.6	126.6	92.9	74.7

FURTHER CAPS AND MEASURES

Table 16: Gain caps of 5 and 20 points per task (same conventions as Table 2, including the inverted-CDF median).

	cap gain5			cap gain20		
rule	feas.	K	L_1	feas.	K	L_1
OLLB (6 tasks), N = 4495 non-leading targets						
mean	13/4495	6†	27.7†	830/4495	5†	85.7†
geometric mean	6/4495	1†	3.7†	959/4495	4†	72.4†
harmonic mean	13/4495	4†	15.8†	1157/4495	4†	62.0†
median	40/4495	1†	4.2†	641/4495	3†	30.4†
min	105/4495	2†	6.2†	2873/4495	3†	33.7†
HELM MMLU (57 subjects), N = 78 non-leading targets						
mean	11/78	34†	168†	56/78	28†	539†
geometric mean	8/78	31†	151†	50/78	25†	479†
harmonic mean	8/78	30†	149†	50/78	25†	488†
median	16/78	9†	16.6†	66/78	19†	76.9†
min	7/78	2†	2.4†	34/78	6†	42.2†
Borda	11/78	24†	56.0†	49/78	44†	331†
Condorcet winner	16/78	23†	55.4†	65/78	29†	169†
plurality	29/78	9†	20.7†	74/78	9†	45.1†
OLLB top-500 frame (6 tasks; ordinal rules), N = 499 non-leading targets						
mean	–	–	–	–	–	–
geometric mean	–	–	–	–	–	–
harmonic mean	–	–	–	–	–	–
median	–	–	–	–	–	–
min	–	–	–	–	–	–
Borda	–	–	–	–	–	–
Condorcet winner	–	–	–	–	–	–
plurality	–	–	–	–	–	–

Table 17: Best-existing-score-plus-2 cap and the empirical 95th-percentile gain scale (descriptive only).

	cap front2			cap emp95		
rule	feas.	K	L_1	feas.	K	L_1
OLLB (6 tasks), N = 4495 non-leading targets						
mean	4495/4495	4 [3, 4]	181 [137, 229]	335/4495	$4^\dagger$	$65.3^\dagger$
geometric mean	4495/4495	5 [4, 5]	158 [117, 207]	111/4495	$5^\dagger$	$60.8^\dagger$
harmonic mean	4495/4495	4 [4, 5]	129 [92.7, 176]	77/4495	$4^\dagger$	$54.2^\dagger$
median	4495/4495	3 [3, 4]	109 [80.2, 151]	141/4495	$2^\dagger$	$18.8^\dagger$
min	4495/4495	3 [3, 5]	43.2 [29.5, 67.8]	65/4495	$2^\dagger$	$5.9^\dagger$
HELM MMLU (57 subjects), N = 78 non-leading targets						
mean	78/78	31 [23, 38]	661 [440, 1057]	–	–	–
geometric mean	78/78	30 [21, 37]	641 [425, 1037]	–	–	–
harmonic mean	78/78	30 [20, 37]	624 [413, 1019]	–	–	–
median	78/78	21 [14, 27]	111 [37.7, 257]	–	–	–
min	78/78	11 [7, 18]	97.5 [45.9, 242]	–	–	–
Borda	78/78	46 [38, 52]	500 [291, 893]	–	–	–
Condorcet winner	78/78	29 [28, 29]	213 [108, 354]	–	–	–
plurality	78/78	9 [9, 9]	46.7 [25.3, 79.8]	–	–	–
OLLB top-500 frame (6 tasks; ordinal rules), N = 499 non-leading targets						
mean	499/499	3 [3, 3]	71.1 [61.6, 82.0]	–	–	–
geometric mean	499/499	3 [2, 3]	50.8 [43.2, 61.0]	–	–	–
harmonic mean	499/499	2 [2, 2]	32.1 [27.2, 37.9]	–	–	–
median	499/499	2 [2, 2]	26.7 [20.6, 34.4]	–	–	–
min	499/499	2 [2, 2]	12.3 [8.4, 19.4]	–	–	–
Borda	499/499	4 [3, 4]	36.4 [27.1, 47.0]	–	–	–
Condorcet winner	499/499	4 [4, 4]	47.0 [39.3, 54.8]	–	–	–
plurality	499/499	2 [2, 2]	26.3 [23.0, 28.6]	–	–	–

Table 18: Largest single-task rise L_∞, same targets and caps.

	cap ceil		cap gain10	
rule	feas.	L_∞	feas.	L_∞
OLLB (6 tasks), N = 4495 non-leading targets				
mean	4495/4495	30.1 [22.8, 38.1]	113/4495	$8.5^\dagger$
geometric mean	4495/4495	27.8 [21.7, 34.8]	142/4495	$8.7^\dagger$
harmonic mean	4495/4495	24.9 [19.7, 31.0]	264/4495	$8.5^\dagger$
median	4495/4495	36.2 [28.5, 44.4]	203/4495	$8.2^\dagger$
min	4495/4495	17.9 [14.6, 21.5]	446/4495	$6.6^\dagger$
HELM MMLU (57 subjects), N = 78 non-leading targets				
mean	78/78	12.6 [8.2, 20.3]	29/78	$6.5^\dagger$
geometric mean	78/78	13.8 [9.0, 21.9]	24/78	$5.6^\dagger$
harmonic mean	78/78	15.2 [10.0, 24.0]	20/78	$5.7^\dagger$
median	78/78	9.8 [5.8, 16.4]	40/78	$5.8^\dagger$
min	78/78	22.2 [13.3, 29.0]	14/78	$4.6^\dagger$
Borda	78/78	15.0 [8.2, 23.9]	25/78	$5.5^\dagger$
Condorcet winner	78/78	10.1 [5.8, 17.4]	38/78	$5.8^\dagger$
plurality	78/78	6.1 [3.5, 10.7]	57/78	$4.9^\dagger$
OLLB top-500 frame (6 tasks; ordinal rules), N = 499 non-leading targets				
mean	499/499	11.8 [10.3, 13.7]	113/499	$8.5^\dagger$
geometric mean	499/499	11.0 [9.8, 12.3]	142/499	$8.7^\dagger$
harmonic mean	499/499	9.9 [8.5, 11.9]	261/499	$8.5^\dagger$
median	499/499	10.9 [8.6, 13.4]	200/499	$8.2^\dagger$
min	499/499	7.6 [5.4, 11.1]	348/499	$6.0^\dagger$
Borda	499/499	8.6 [6.4, 11.7]	325/499	$6.9^\dagger$
Condorcet winner	499/499	14.4 [12.6, 18.0]	60/499	$7.1^\dagger$
plurality	499/499	15.2 [13.4, 16.8]	53/499	$8.6^\dagger$

Table 19: Targets of original mean rank 2–10 only.

	cap ceil			cap gain10		
rule	feas.	K	L_1	feas.	K	L_1
OLLB (6 tasks), N = 9 non-leading targets						
mean	9/9	1 [1, 1]	24.6 [7.9, 27.7]	9/9	3 [1, 3]	24.6 [7.9, 27.7]
geometric mean	9/9	1 [1, 1]	21.3 [3.7, 22.1]	9/9	3 [1, 3]	22.7 [3.7, 24.3]
harmonic mean	9/9	1 [1, 1]	17.7 [2.0, 19.8]	9/9	2 [1, 3]	17.8 [2.0, 22.8]
median	9/9	1 [1, 1]	0.8 [0.3, 8.4]	9/9	1 [1, 1]	0.8 [0.3, 8.4]
min	9/9	2 [1, 2]	12.0 [3.7, 14.6]	7/9	$2^{\dagger}$	$6.2^{\dagger}$
HELM MMLU (57 subjects), N = 9 non-leading targets						
mean	9/9	4 [1, 5]	156 [41.8, 177]	9/9	16 [5, 18]	156 [41.8, 177]
geometric mean	9/9	4 [2, 4]	139 [38.3, 164]	9/9	15 [4, 18]	144 [38.3, 174]
harmonic mean	9/9	4 [2, 4]	128 [38.0, 155]	9/9	14 [4, 18]	138 [38.0, 177]
median	9/9	4 [2, 8]	3.3 [0.4, 13.5]	9/9	4 [2, 8]	3.3 [0.4, 13.5]
min	9/9	2 [1, 3]	3.0 [2.3, 23.6]	6/9	$1^{\dagger}$	$2.3^{\dagger}$
Borda	9/9	16 [10, 19]	54.0 [21.1, 56.5]	9/9	19 [10, 21]	54.0 [21.1, 56.5]
Condorcet winner	9/9	16 [8, 18]	20.7 [4.6, 29.3]	9/9	16 [8, 18]	20.7 [4.6, 29.3]
plurality	9/9	7 [4, 8]	5.0 [1.4, 7.7]	9/9	7 [4, 8]	5.0 [1.4, 7.7]
OLLB top-500 frame (6 tasks; ordinal rules), N = 9 non-leading targets						
mean	9/9	1 [1, 1]	24.6 [7.9, 27.7]	9/9	3 [1, 3]	24.6 [7.9, 27.7]
geometric mean	9/9	1 [1, 1]	21.3 [3.7, 22.1]	9/9	3 [1, 3]	22.7 [3.7, 24.3]
harmonic mean	9/9	1 [1, 1]	17.7 [2.0, 19.8]	9/9	2 [1, 3]	17.8 [2.0, 22.8]
median	9/9	1 [1, 1]	0.8 [0.3, 8.4]	9/9	1 [1, 1]	0.8 [0.3, 8.4]
min	9/9	2 [1, 2]	12.0 [3.7, 14.6]	7/9	$2^{\dagger}$	$6.2^{\dagger}$
Borda	9/9	1 [1, 1]	3.8 [1.6, 5.0]	9/9	1 [1, 1]	3.8 [1.6, 5.0]
Condorcet winner	9/9	2 [2, 3]	10.0 [6.7, 11.9]	9/9	3 [3, 3]	10.0 [6.7, 11.9]
plurality	9/9	2 [2, 2]	7.2 [6.7, 10.5]	7/9	$2^{\dagger}$	$7.2^{\dagger}$

TOP-500 FRAME, MARGIN, RIVAL SET AND SOLVER AUDIT

Table 20: Top-500 Open LLM Leaderboard frame (6 tasks), all rules including the ordinal ones; 499 non-leading targets. Conventions of Table 2.

	cap ceil			cap gain10		
rule	feas.	K	L_1	feas.	K	L_1
OLLB top-500 frame (6 tasks; ordinal rules), N = 499 non-leading targets						
mean	499/499	1 [1, 1]	71.1 [61.6, 82.0]	113/499	$6^{\dagger}$	$50.9^{\dagger}$
geometric mean	499/499	1 [1, 2]	48.5 [42.1, 56.4]	142/499	$5^{\dagger}$	$48.2^{\dagger}$
harmonic mean	499/499	2 [1, 2]	31.9 [27.1, 37.2]	261/499	$5^{\dagger}$	$39.4^{\dagger}$
median	499/499	2 [2, 2]	26.5 [20.5, 34.4]	200/499	$3^{\dagger}$	$18.3^{\dagger}$
min	499/499	2 [2, 2]	12.3 [8.4, 19.4]	348/499	$2^{\dagger}$	$9.6^{\dagger}$
Borda	499/499	4 [3, 4]	36.4 [27.1, 47.0]	325/499	$4^{\dagger}$	$30.6^{\dagger}$
Condorcet winner	499/499	4 [4, 4]	47.0 [39.3, 54.8]	60/499	$4^{\dagger}$	$23.3^{\dagger}$
plurality	499/499	2 [2, 2]	26.3 [23.0, 28.6]	53/499	$2^{\dagger}$	$13.1^{\dagger}$

Table 21: Margin sensitivity: $\varepsilon = 0.001$ against 0.1 for the 20 non-leading targets of mean rank 2–21 per frame (not a census).

dataset	cap	rule	feasibility changed	K changed (max $\lvert\Delta K\rvert$)	L_1 increase, median (max)
HELM MMLU (57 subjects)					
	ceiling	mean	0/20	2/20 (1)	5.64 (5.64)
	ceiling	Borda	0/20	0/20 (0)	5.05 (5.64)
	+10	mean	0/20	7/20 (1)	5.64 (5.64)
	+10	Borda	0/20	1/20 (1)	5.05 (5.50)
OLLB, full frame (6 tasks)					
	ceiling	mean	0/20	0/20 (0)	0.59 (0.59)
	+10	mean	0/20	2/20 (1)	0.59 (0.59)
OLLB, top-500 frame (6 tasks)					
	ceiling	mean	0/20	0/20 (0)	0.59 (0.59)
	ceiling	Borda	0/20	0/20 (0)	0.07 (0.23)
	+10	mean	0/20	2/20 (1)	0.59 (0.59)
	+10	Borda	0/20	0/20 (0)	0.07 (0.15)

Table 22: Competitor-frame dependence of K under the 10-point cap for ten fixed targets (mean rank 2–11): the mean's leader removed, and ten random half-frames. A fixed panel, descriptive. The ceiling-cap version is Table 23.

		original mean leader removed			ten random-half frames, median [2.5–97.5%]		
rule	reachable originally	reachable	K lower	median ΔK	reachable	K lower	median ΔK
HELM MMLU (57 subjects)							
mean	10	10	5	−1	10 [10, 10]	5 [0, 10]	−1 [−15, 0]
geometric mean	10	10	0	0	10 [10, 10]	0 [0, 9]	0 [−15, 0]
harmonic mean	10	10	0	0	10 [10, 10]	0 [0, 9]	0 [−14, 0]
median	10	10	1	0	10 [10, 10]	1 [0, 7]	0 [−1, 0]
min	7	7	0	0	7 [7, 9]	0 [0, 8]	0 [−2, 0]
Borda	10	10	10	−3	10 [10, 10]	10 [1, 10]	−5 [−18, 0]
Condorcet winner	10	10	7	−1	10 [10, 10]	7 [1, 10]	−2 [−12, 0]
plurality	10	10	0	1	10 [10, 10]	0 [0, 8]	2 [−4, 5]
OLLB, full frame (6 tasks)							
mean	10	10	3	0	10 [10, 10]	0 [0, 7]	0 [−1, 0]
geometric mean	10	10	5	−1	10 [10, 10]	0 [0, 8]	0 [−1, 0]
harmonic mean	10	10	5	−1	10 [10, 10]	0 [0, 6]	0 [−1, 0]
median	10	10	0	0	10 [10, 10]	0 [0, 3]	0 [0, 0]
min	7	7	0	0	8 [7, 8]	1 [0, 1]	0 [0, 0]
OLLB, top-500 frame (6 tasks)							
mean	10	10	3	0	10 [10, 10]	3 [0, 10]	0 [−2, 0]
geometric mean	10	10	5	−1	10 [10, 10]	5 [0, 10]	−1 [−1, 0]
harmonic mean	10	10	5	−1	10 [10, 10]	5 [0, 8]	−1 [−1, 0]
median	10	10	0	0	10 [10, 10]	0 [0, 5]	0 [−1, 0]
min	7	7	0	0	7 [7, 8]	0 [0, 1]	0 [0, 0]
Borda	10	10	0	0	10 [10, 10]	1 [0, 5]	0 [−1, 0]
Condorcet winner	10	10	1	0	10 [10, 10]	1 [0, 6]	0 [−1, 0]
plurality	8	8	1	0	8 [3, 10]	2 [0, 5]	0 [0, 0]

Table 23: As Table 22 under the ceiling cap.

		original mean leader removed			ten random-half frames, median [2.5–97.5%]		
rule	reachable originally	reachable	K lower	median ΔK	reachable	K lower	median ΔK
HELM MMLU (57 subjects)							
mean	10	10	2	0	10 [10, 10]	2 [0, 10]	0 [−4, 0]
geometric mean	10	10	0	0	10 [10, 10]	0 [0, 9]	0 [−3, 0]
harmonic mean	10	10	0	0	10 [10, 10]	0 [0, 9]	0 [−3, 0]
median	10	10	1	0	10 [10, 10]	1 [0, 7]	0 [−1, 0]
min	10	10	0	0	10 [10, 10]	0 [0, 8]	0 [−2, 0]
Borda	10	10	10	−3	10 [10, 10]	10 [2, 10]	−4 [−15, 0]
Condorcet winner	10	10	7	−1	10 [10, 10]	7 [1, 10]	−2 [−12, 0]
plurality	10	10	0	1	10 [10, 10]	0 [0, 8]	2 [−4, 5]
OLLB, full frame (6 tasks)							
mean	10	10	1	0	10 [10, 10]	0 [0, 3]	0 [0, 0]
geometric mean	10	10	1	0	10 [10, 10]	0 [0, 3]	0 [0, 0]
harmonic mean	10	10	1	0	10 [10, 10]	0 [0, 2]	0 [0, 0]
median	10	10	0	0	10 [10, 10]	0 [0, 3]	0 [0, 0]
min	10	10	0	0	10 [10, 10]	0 [0, 0]	0 [0, 0]
OLLB, top-500 frame (6 tasks)							
mean	10	10	1	0	10 [10, 10]	1 [0, 4]	0 [0, 0]
geometric mean	10	10	1	0	10 [10, 10]	1 [0, 3]	0 [0, 0]
harmonic mean	10	10	1	0	10 [10, 10]	1 [0, 3]	0 [0, 0]
median	10	10	0	0	10 [10, 10]	0 [0, 5]	0 [−1, 0]
min	10	10	0	0	10 [10, 10]	0 [0, 0]	0 [0, 0]
Borda	10	10	0	0	10 [10, 10]	1 [0, 4]	0 [0, 0]
Condorcet winner	10	10	1	0	10 [10, 10]	1 [0, 3]	0 [0, 0]
plurality	10	10	1	0	10 [10, 10]	1 [0, 3]	0 [0, 0]

Table 24: Borda solver audit: dynamic program with witness check against the exact MILP fallback, HELM and top-500 instances. Audit counts computed after the main runs, not telemetry of them.

dataset	cap	instances	initial winner	DP accepted	DP infeasible	MILP fallback
HELM MMLU	ceiling	158	2	156	0	0
HELM MMLU	front0	158	2	156	0	0
HELM MMLU	front2	158	2	156	0	0
HELM MMLU	+5	158	2	22	134	0
HELM MMLU	+10	158	2	50	106	0
HELM MMLU	+20	158	2	98	58	0
OLLB top-500	ceiling	1 000	2	998	0	0
OLLB top-500	front0	1 000	2	998	0	0
OLLB top-500	front2	1 000	2	998	0	0
OLLB top-500	+10	1 000	2	650	348	0
total		4 948	20	4 282	646	0

C PROOFS OF THE EXACT RESULTS

Setting. Fix a target with scores $x \in [0, 100]^n$ and rival rows $r_1, \dots, r_m$, which stay fixed. The scenario cap on task t gives the effective cap $c_t = \max(x_t, \text{scenario cap}_t) \le 100$ and headroom $h_t = c_t - x_t \ge 0$; admissible plans are $y \in \prod_t [x_t, c_t]$, so $L_1 = \sum_t (y_t - x_t)$, $L_\infty = \max_t (y_t - x_t)$ and $K = |\{t : y_t > x_t\}|$. For a score rule f let $\tau = \max_j f(r_j)$ be the best rival value; the target is first iff $f(y) \ge \tau + \varepsilon$ (a target with $f(x) \ge \tau + \varepsilon$, or one already uniquely first, costs 0 by convention, and the formulas below concern the other targets). Since $\varepsilon > 0$ and all rules below are continuous in y, the set of first-placing plans is closed, so every minimum is attained. Write $(u)^+ = \max(0, u)$.

Proposition 1 (Mean)**.** *Let $g = \tau + \varepsilon - \bar{x} > 0$. The target can become first iff $\sum_t h_t \ge ng$. Then the minimum L_1 is ng, the minimum K is the least k such that the k largest headrooms sum to at least ng, and the minimum L_∞ is the least d with $\sum_t \min(d, h_t) \ge ng$.*

Proof. $\bar{y} \ge \tau + \varepsilon$ iff $\sum_t (y_t - x_t) \ge ng$, and $\sum_t (y_t - x_t) = L_1$ because $y \ge x$. Hence $L_1 \ge ng$ for every first-placing plan, and any plan with $y_t - x_t \in [0, h_t]$ and sum exactly ng attains it; such a plan exists iff $\sum_t h_t \ge ng$, which is the feasibility condition. A plan touching a set S of tasks raises the sum by at most $\sum_{t \in S} h_t$, which is at most the sum of the $|S|$ largest headrooms; conversely any amount in $[0, \sum_{t \in S} h_t]$ is attainable on S, so the least $|S|$ is the stated prefix length. A plan with all rises at most d gains at most $\sum_t \min(d, h_t)$, and every amount up to that is attainable, giving the L_∞ statement. □

Proposition 2 (Minimum)**.** *Let $b = \tau + \varepsilon$ with τ the best rival minimum. The target can become first iff $c_t \ge b$ for all t. Then $L_1 = \sum_t (b - x_t)^+$, $K = |\{t : x_t < b\}|$ and $L_\infty = \max_t (b - x_t)^+$, each the minimum of its measure.*

Proof. $\min_t y_t \ge b$ iff $y_t \ge b$ for every t. For t with $x_t \ge b$ this holds with $y_t = x_t$ (and then $c_t \ge x_t \ge b$). For t with $x_t < b$ we need $y_t \ge b$, possible iff $c_t \ge b$, and the cheapest such value is $y_t = b$. The three measures are separable over tasks (sum, count, maximum) and each coordinate is forced to at least $\max(x_t, b)$, so the stated values are lower bounds attained by $y_t = \max(x_t, b)$. One coordinate with $c_t < b$ blocks first place regardless of the other coordinates. □

Proposition 3 (Odd median)**.** *Let $n = 2q - 1$ and $b = \tau + \varepsilon$ with τ the best rival median. Let $a_t = (b - x_t)^+$ if $c_t \ge b$ and $a_t = \infty$ otherwise, and $A = |\{t : x_t \ge b\}|$. The target can become first iff at least q coordinates have $c_t \ge b$. Then the minimum L_1 is the sum of the q smallest a_t, the minimum L_∞ is the qth smallest a_t, and the minimum K is $(q - A)^+$.*

Proof. For odd n the median is the qth largest coordinate, so $\mathrm{med}(y) \ge b$ iff at least q coordinates satisfy $y_t \ge b$. Such a coordinate t costs at least a_t (and is impossible if $a_t = \infty$) in each of the three measures: $y_t - x_t \ge (b - x_t)^+$. For any first-placing plan let S be a set of q coordinates with $y_t \ge b$. Then $L_1 \ge \sum_{t \in S} a_t \ge$ the sum of the q smallest a_t, and $L_\infty \ge \max_{t \in S} a_t \ge$ the qth smallest a_t; raising exactly the q cheapest coordinates to $\max(x_t, b)$ attains both. For K, the A coordinates already at least b cost nothing and count toward q; each touched coordinate adds at most one new coordinate at least b, so $K \ge (q - A)^+$, attained by raising $(q - A)^+$ coordinates with $x_t < b \le c_t$ to b, which is possible whenever the plan is feasible. □

Proposition 4 (Even median). *Let $n = 2q$ and $b = \tau + \varepsilon$. A plan makes the target first iff there is a threshold $a \geq b$ such that at least q coordinates are at least a and at least $q + 1$ are at least $2b - a$. Minimum L_1 and L_∞ over plans are attained at a threshold a in the finite set $\{b\} \cup \{x_t, c_t, 2b - x_t, 2b - c_t\}$, and the minimum K is found by enumerating pairs of thresholds from the values $\{x_t, c_t\}$.*

Proof. Order y decreasingly as $z_1 \geq \cdots \geq z_n$; the median is $(z_q + z_{q+1})/2$, which is at least b iff $z_q + z_{q+1} \geq 2b$. Given such a plan take $a = z_q \geq b$; then q coordinates are $\geq a$ and $z_{q+1} \geq 2b - a$ gives $q + 1$ coordinates $\geq 2b - a$. Conversely such an a gives $z_q \geq a$ and $z_{q+1} \geq 2b - a$, whose sum is at least $2b$. For a fixed a the cheapest plan lifts a set Q of q coordinates to a and one further coordinate e to $2b - a$ (costs $(a - x_t)^+$ and $(2b - a - x_e)^+$, impossible if the cap is lower). For a fixed assignment (Q, e) the cost is piecewise linear in a with breakpoints at x_t and $2b - x_e$, over the interval of a where all caps are respected, whose endpoints are b, some c_t or some $2b - c_t$. A piecewise-linear function on an interval attains its minimum at a breakpoint or an endpoint, so enumerating the listed values and optimizing the assignment (a sort) for each yields the optimum; L_∞ is the same with the maximum in place of the sum. For K, raising a touched coordinate to its cap never hurts, so a plan is a touched set with $y_t = c_t$ on it and $y_t = x_t$ elsewhere; its middle pair takes values in $\{x_t, c_t\}$, and for each threshold pair from these values the number of coordinates that must be touched is a simple count, so the minimum over pairs is the minimum K. These solvers are compared with exhaustive linear programs in `analysis/tests/test_scorerules.py`. □

Proposition 5 (Cap threshold). *Let f be a rule for which the set of first-placing plans in the ceiling box is closed (all score rules above, and the ordinal rules on their finite level sets). Let d^* be the minimum L_∞ over first-placing plans under the ceiling cap, attained by closedness. Under the cap "at most $+h$ points per task" the target can become first iff $d^* \leq h$.*

Proof. The gain-h box has effective caps $c_t = \max(x_t, \min(x_t + h, 100)) = \min(x_t + h, 100)$, so it equals $\{y : x \leq y \leq 100,\ y_t - x_t \leq h\}$, the set of ceiling-box plans with $L_\infty \leq h$. A first-placing plan exists in it iff some first-placing ceiling plan has $L_\infty \leq h$, iff $d^* \leq h$. No monotonicity of f is needed. □

Proposition 6 (Finite levels for ordinal rules). *The Borda total, the plurality count and the pairwise task-majority relations that define the Condorcet requirement depend on the plan y only through the comparisons of each y_t with the rival scores $s_{r,t}$ on that task (below, equal, above). Let $\mathcal{L}$ be the set of plans in which every y_t is the original x_t, equals a rival score $s_{r,t} \geq x_t$, or equals $s_{r,t} + \delta$ for such a rival score (δ the margin step). Each cost measure is minimised over $\mathcal{L}$ by an integer linear program. The minimum over $\mathcal{L}$ is an upper bound on the infimum over real plans, and first place being unreachable on $\mathcal{L}$ does not prove it unreachable over the reals.*

Proof. A comparison outcome is constant on each open interval between consecutive distinct values of $\{x_t\} \cup \{s_{r,t}\}$ and on each such value. Within an outcome class the cheapest coordinate value is the lower end of the class; for "strictly above s" that end is an infimum that is not attained, which is why the margin step $\delta > 0$ is a convention that makes a strict pass attainable. Values below x_t are inadmissible and rivals scoring below x_t are already beaten on that task, so only rival scores at or above x_t matter. Every plan in $\mathcal{L}$ is a real admissible plan, hence the minimum over $\mathcal{L}$ is at least the infimum over real plans. With one-hot variables $z_{t,\ell}$ selecting a level of each task, the Borda total, the plurality count and each pairwise task-majority indicator are linear in z (a rival's loss is the sum of the target's gains against it), so the costs are minimised by integer linear programming. The dynamic program of Section 4 solves the relaxation that keeps only the constraint against the rival of highest original total; if its witness satisfies all rival constraints it is optimal for the full problem, and infeasibility of the relaxation proves infeasibility on $\mathcal{L}$. □

Proposition 7 (Constructed counterexamples). *The three rows of Table 1 are exact.*

Proof. By Propositions 1–3 with $n = 5$, $q = 3$. Row 1 (target $(10, 0, 0, 0, 0)$, rival $(3, 3, 3, 3, 3)$, cap 100): the mean needs 5.005 points, available on the first task (headroom 90), so $K = 1$; the median threshold is $b = 3.001$ and the first coordinate already exceeds it, so two more must be lifted ($K = 2$, $L_1 = 6.002$); the minimum needs all four others at b ($K = 4$, $L_1 = 12.004$). Row 2 (zero target, rival $(9, 9, 1, 1, 1)$, gain 5): the mean needs $5 \times 4.201 = 21.005$ points at 5 per task, so

$K = 5$; the median and minimum thresholds are 1.001, so $K = 3$ ($L_1 = 3.003$) and $K = 5$ (5.005). Row 3 (target $(4, 4, 4, 4, 0)$, rival $(9, 9, 1, 1, 1)$, caps $(5.1, 5.1, 5.1, 5.1, 1.1)$): the mean needs 5.005 points at 1.1 per task, so $K = 5$; the target's median 4 already exceeds the rival's 1 (cost 0); the minimum needs only the last task lifted to 1.001 ($K = 1$). By K the order is mean $<$ median $<$ minimum in row 1 but median $<$ minimum $<$ mean in row 3, and the cheapest rule is the mean in row 1 and the median in rows 2 and 3, so the ordering reverses. □

D Data provenance and licences

HELM MMLU: release v1.13.0 (summary dated 2025-01-10) from the public HELM bucket; the 57 per-subject tables form the matrix. **Open LLM Leaderboard `contents`:** Hugging Face dataset commit `9c09a7cae43334062a82cb164f2ef255013dafa2` (2025-03-20), parquet SHA-256 in Table 25. **OLLB `results`:** commit `aa81ecc38fdc5708254b833923368970efdf5ef5` (2025-03-15), used from one JSON only, for approximate per-task item-count scales in the noise sensitivity. Downloads were made on 2026-11-04. No separate licence for the score data is declared by either source (the HELM code is Apache-2.0; the `Hub License` column of OLLB is a model licence, not a data licence); we claim no licence over the scores, redistribute only derived matrices with checksums for verification, and urge users to check current terms. **Reserved-material log:** the analysis position ran no literature search, and neither it nor the writing position sought or opened the reserved prior study or any page reporting it; literature searches for the related-work section used generic terms (social choice, bribery, benchmark aggregation and contamination). No such material was knowingly encountered; none is cited.

Table 25: Full SHA-256 values of the inputs (as in `analysis/data/processed/manifest.json`, `analysis/data/provenance/` and `sha256sum` of the processed files).

OLLB `contents` parquet:
`f13a7061e63133bc8745d89f8c7a56148584940ede22d3b27e21cad1e078dca5`
HELM v1.13.0 release manifest (303 files):
`830bdb10a5518c50689d60c86623cc3d79ff5317d7e6b96408ea7dc2058de12b`
`helm_mmlu57.csv` (HELM matrix):
`fae6683b2e42d4b02116716c7132f6b1bc33cc9db44fd283232e1d397b7c73c6`
`ollb6.csv` (OLLB matrix):
`3bf058715d37613b96b2fb253d8f21cbb49b7129989559eb2eae1b7254a6a999`
`ollb6_lowest_average.csv` (sensitivity frame):
`940ab0937673f0ed4c2955b9701cbbf0f72ece602ab1a3eab9d73506e7bb5798`

E Reproduction: commands and measured times

From `analysis/`, with the pinned `requirements.lock` (Python 3.11.2) and GNU make 4.4.1: `make data` (re-download and verify the public inputs against the SHA-256 values above), `make all` (tests, all frames, caps, sensitivity and control panels, validation, summaries; `make quick` is only a partial run of the HELM rules and the Open LLM Leaderboard score rules), and, for the review and final additions alone, `make review-checks tables` and `make final-additions`. A fresh isolated export is recomputed and compared row by row (absolute tolerance 10^{-5}, infinity and NaN statuses matched) with `src/clean_reproduce.py`, `src/clean_review_reproduce.py` and `src/clean_final_reproduce.py`. Table 26 lists the measured times on two CPUs. The original 16-file study (the seven main and sensitivity files and the nine robustness and control files; 457,772 rows) was recomputed from an initially empty export (`src/clean_reproduce.py`) and compared at absolute tolerance 10^{-5} with matching infinity and NaN statuses: it passed, with a maximum finite difference of 5.8×10^{-9} (HELM main) and zero difference in all 174,000 robustness and control rows (`analysis/results/clean_reproduction.json`). The first attempt stopped at a 120 s proof limit of the mixed-integer solver in one top-500 Condorcet stage (a timeout, not a mismatch and not proved infeasibility; no timeout is mapped to infinity); the same export was resumed with

a 1200 s limit, which changes no constraint or accepted status. The total of 14,758.4 s wall and 12,333.1 s CPU includes the failed stage and excludes idle and setup gaps; it is not an uninterrupted run. This pass uses the bundled pinned processed matrices; re-downloading and verifying the raw sources (`make data`) is a separate step that this recomputation did not repeat. The review and final additions were recomputed separately, each with zero numerical difference.

Table 26: Measured wall-clock and CPU seconds of completed clean recomputations (source: `analysis/results/clean_*runtime.json`); the four original-study rows sum to 14,758.4 s wall and 12,333.1 s CPU.

Clean recomputation	wall (s)	CPU (s)
Original study: validation and tests	37.5	36.4
Original study: main stage, failed at 120 s proof limit	11,443.0	9,398.9
Original study: main stage resumed at 1200 s limit	1,596.5	1,460.5
Original study: robustness, controls, tables	1,681.5	1,437.3
Frame/margin/Borda-audit panels (all three scripts)	452.6	372.6
Target bootstrap (R=1000, seed 23)	13.3	13.1
Headline ranges and shares	12.6	9.0
All-target controls (110,876 records)	278.0	165.3
Controls summary (300 ratio rows)	20.8	18.8
Processing choices (134,860 records)	217.4	137.9
Processing summary	4.8	4.5
Matched frames (250,848 costs, 201 frames)	5066.3	3185.0
Matched-frame summaries	10.6	10.5
Cost units and caps (234,744 records)	3849.9	3673.7
Cost-unit summaries	5.3	5.2